

CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Kameneva

Generated on 2026-10-09 16:31

Run		Perturbations (perturbation_simulation)
Split	train	✓ P1 KO CHGA (50%)
Stimulus penalisation	0.2	✓ P2 KO CHGA
Prior penalisation	1.0	✓ P3 OV STMN2
Genes	97	✓ P4 KO CHGA · OV STMN2
Stimuli	1	
Observed cells	1260	
Timepoints	0, 8, 32, 48, 64, 80	
Cell types	precursors, intermediate, sympathoblasts, chromaffin	
Network shown	inter_simul.npy, network #0 / 1	
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (observed / depth factor, model draws at the reference depth)	

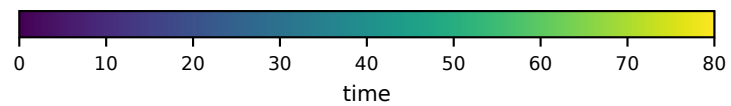
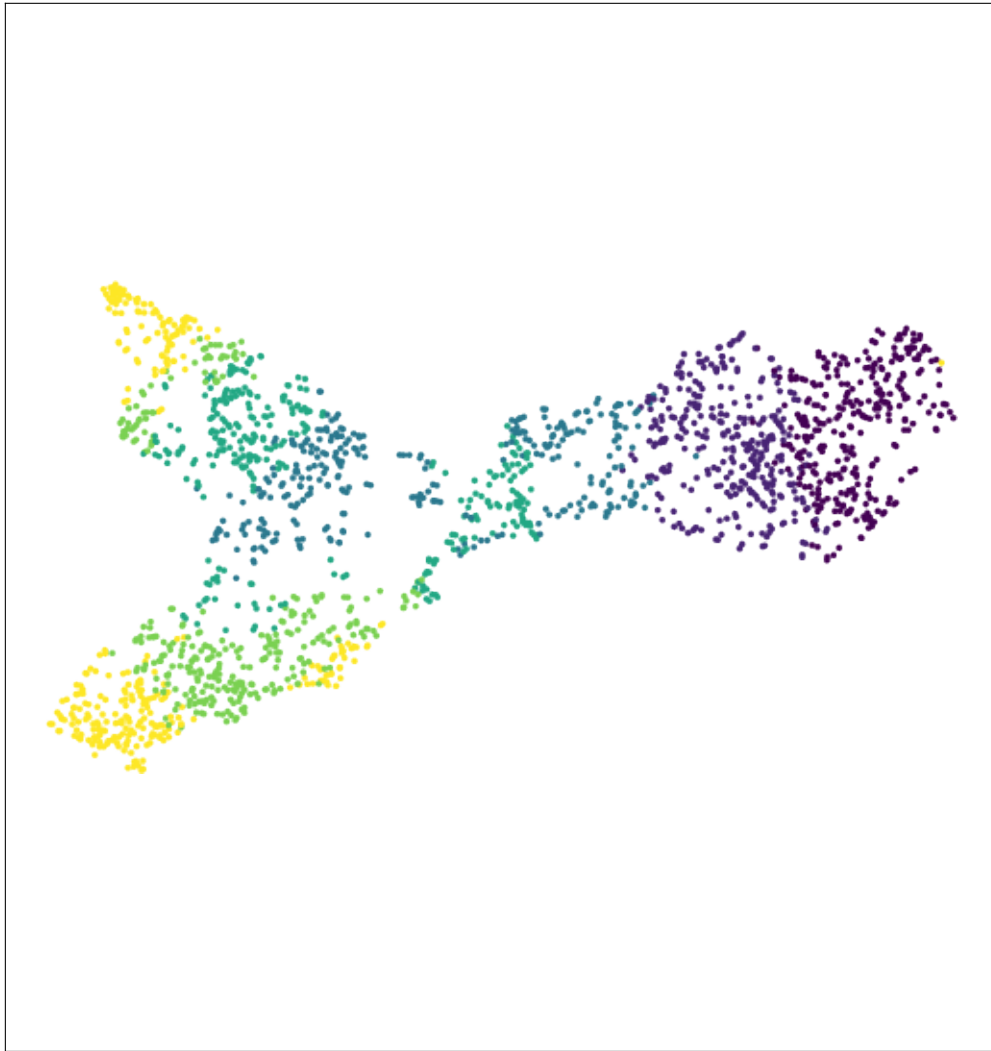
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 2. Gene regulatory network — regulatory power (violin plots), top-20 regulators and top-20 regulated genes (mutual top-10 edges)
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 4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
 5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
 6. Held-out test cells — predictions with the network fixed vs the test data

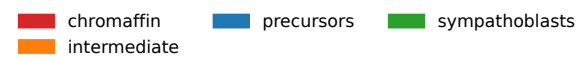
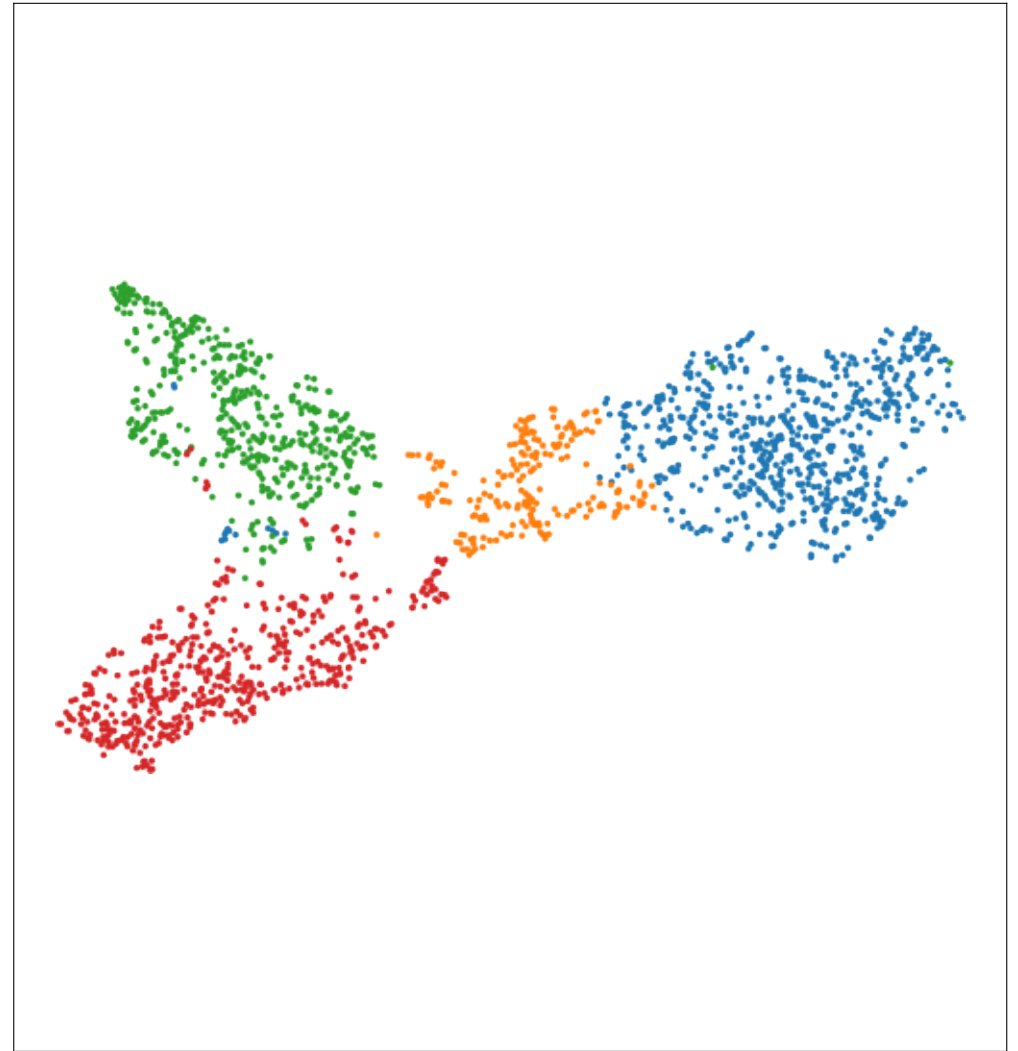
Data — UMAP of all the cells

Data/data.h5ad, 1800 cells: log1p(counts), 97 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.

colour = time



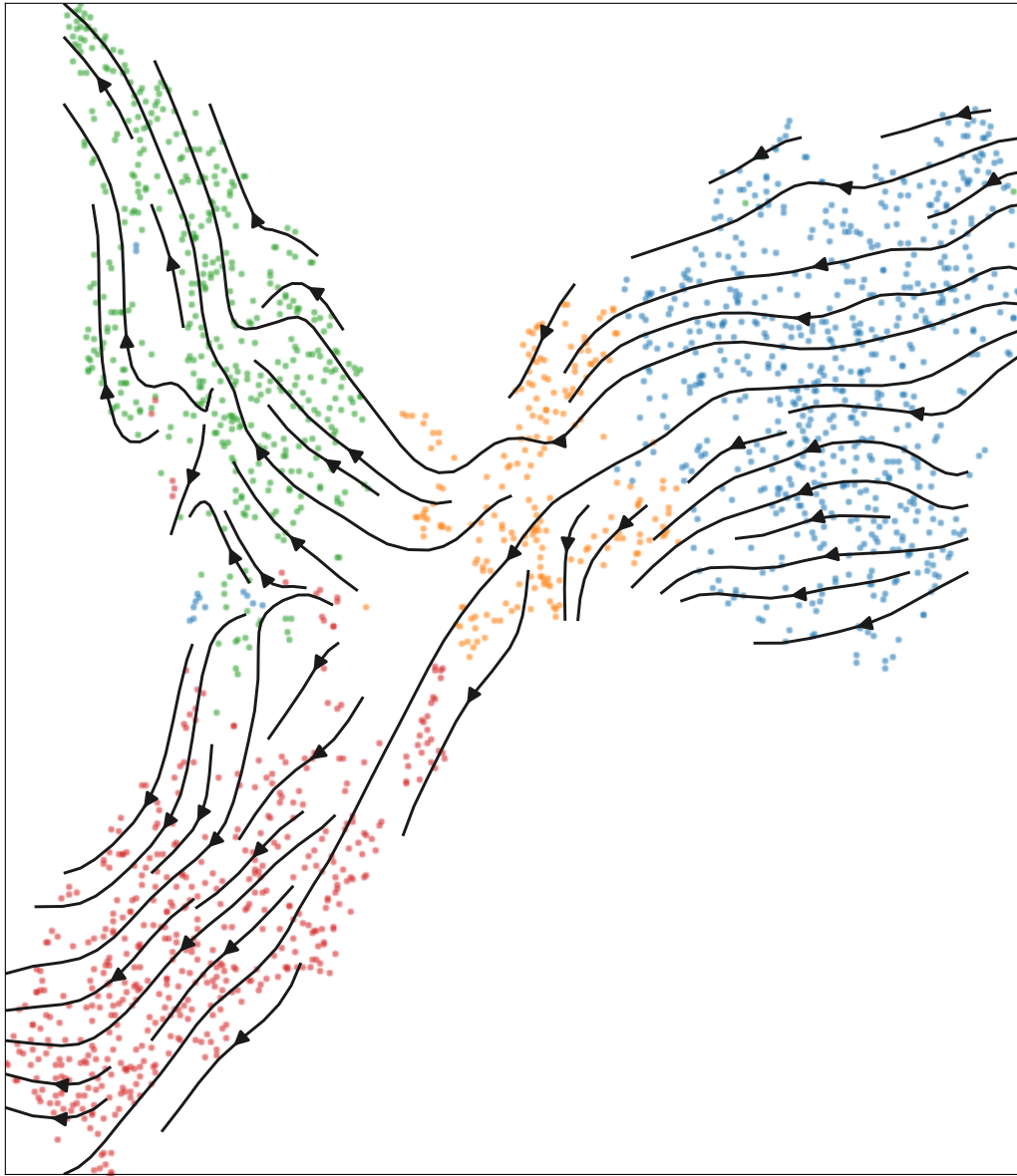
colour = cell type



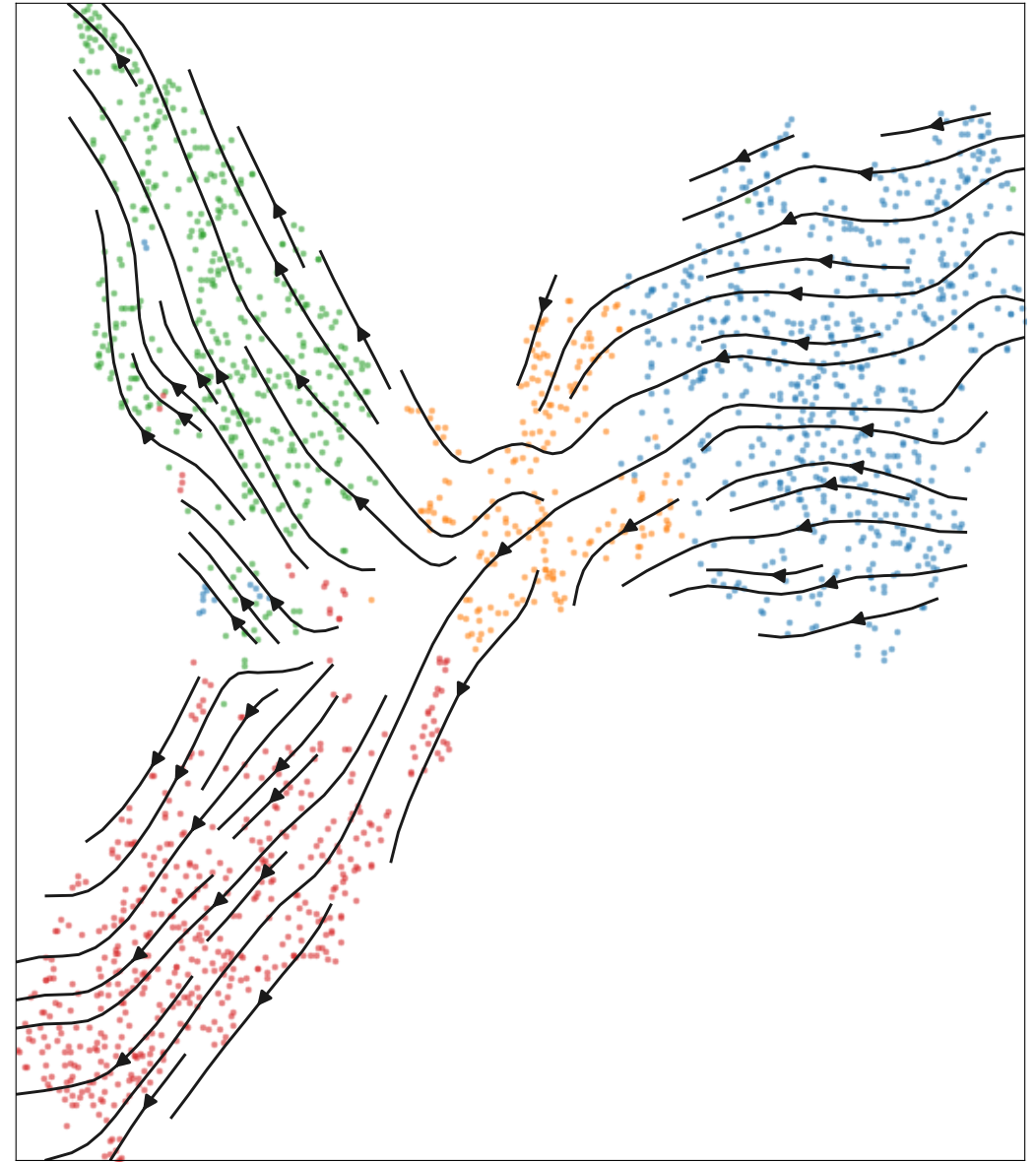
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant ($x(t+1) - x(t)$, not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.68 (PCA space). CardamomOT: final soft couplings of the trajectories (1260 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.

Classical OT, colour = cell type



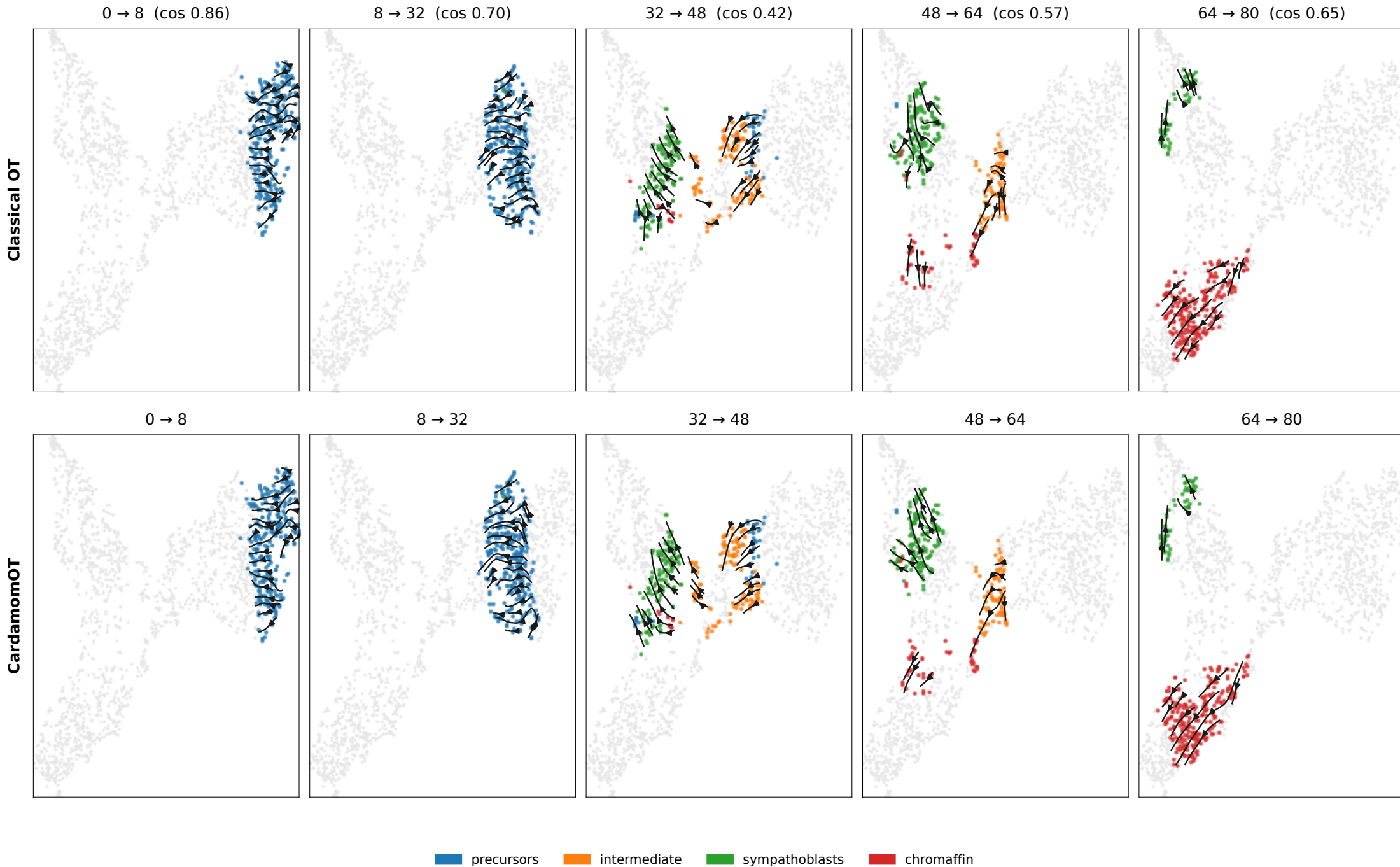
CardamomOT (final soft couplings), colour = cell type



precursors intermediate sympathoblasts chromaffin

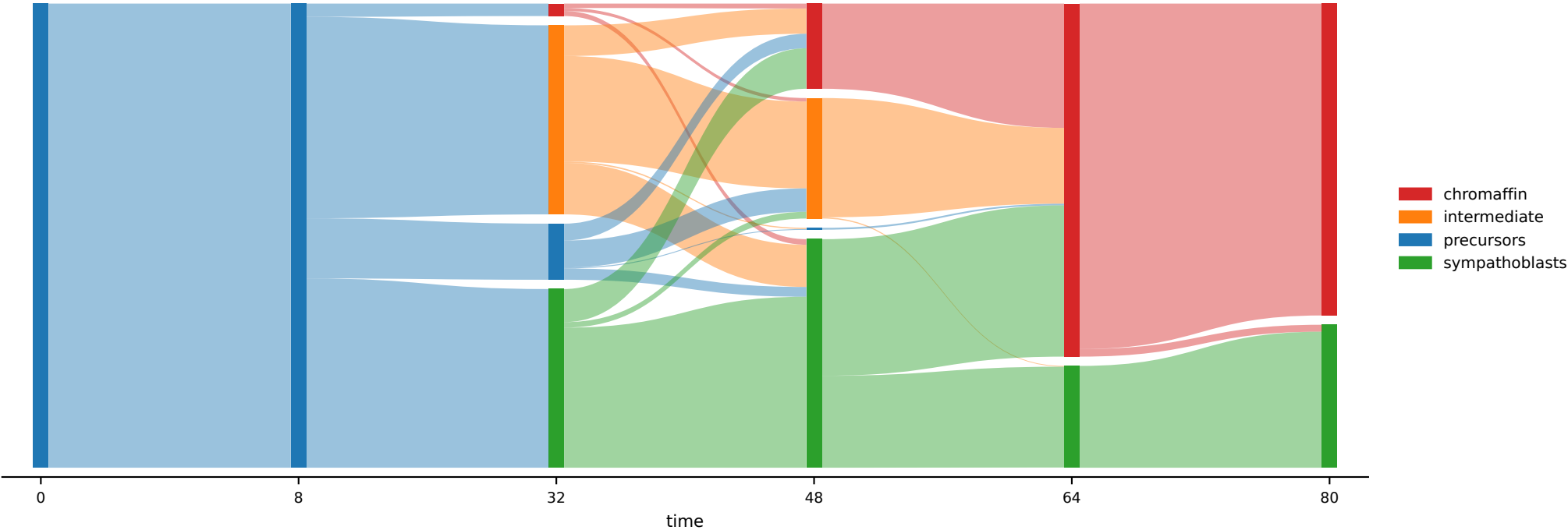
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_{from} (coloured; other cells in grey) towards their mean descendant at t_{to} , kNN projection among the cells of t_{from} and t_{to} . cos: weighted cosine between the two methods on the cells of t_{from} with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave $\propto$ fate and arrive $\propto$ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



Top-10 genes up in the ancestors (vs ancestors of the other cells)

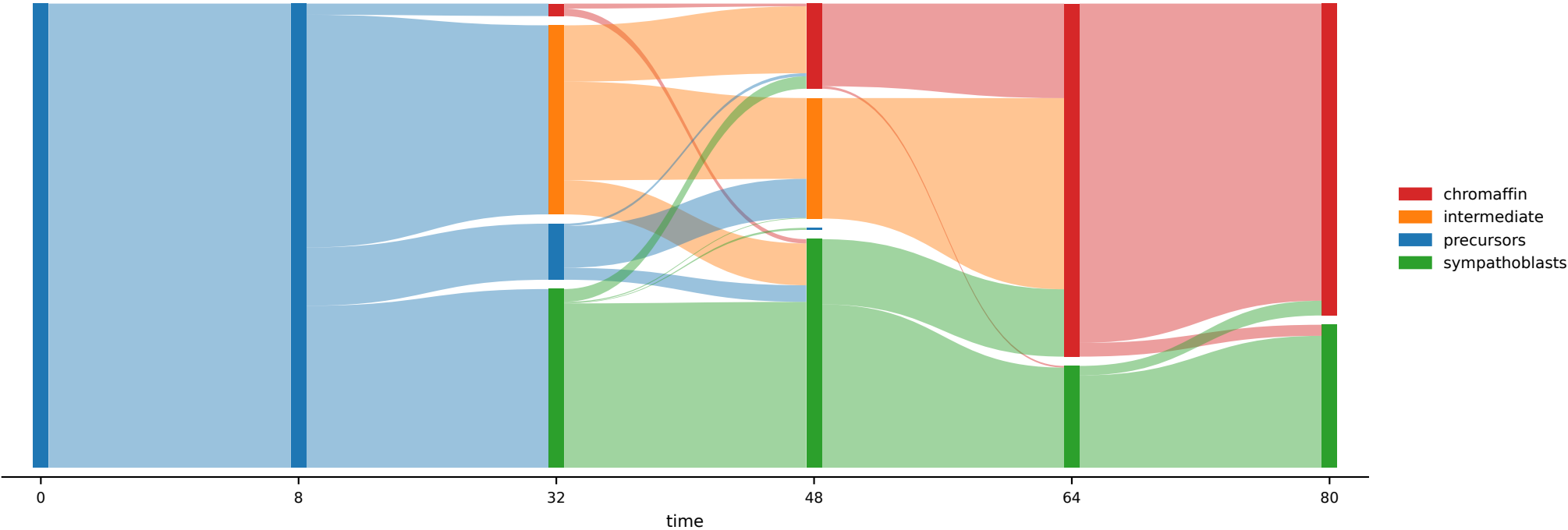
Top-10 genes up in the cell type (vs the other cells)

chromaffin	CHGA	CHGB	MEG3	DLK1	PCSK1N	TH	SCG2	RAMP1	DBH	RGS4
intermediate	PLP1	PTPRZ1	ZEB2	POSTN	COL5A2	QKI	EDNRB	SPARC	VIM	MPZ
precursors	S100B	VIM	METRN	S100A10	PLP1	SPARC	ANXA2	EDNRB	MPZ	LGALS1
sympathoblasts	PTMA	GAP43	TMSB10	STMN2	RTN1	MAP1B	BASP1	GAL	UCHL1	MDK

CHGA	DLK1	CHGB	RAMP1	TH	SCG2	HTATSF1	PCSK1N	PENK	DBH
MT-CYB	POSTN	ZEB2	COL1A1	COL5A2	VCAN	ALDH1A1	PLP1	OLFML2A	NRXN1
PLP1	S100B	VIM	SPARC	EDNRB	PTPRZ1	MPZ	METRN	COL5A2	IFITM3
STMN2	RTN1	BASP1	GAP43	MAP1B	GAL	PHOX2A	UCHL1	CD24	TMSB10

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave $\propto$ fate and arrive $\propto$ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



Top-10 genes up in the ancestors (vs ancestors of the other cells)

Top-10 genes up in the cell type (vs the other cells)

chromaffin	CHGA CHGB DLK1 SCG2 TH PENK RAMP1 PCSK1N DBH RGS4	CHGA DLK1 CHGB TH MEG3 RAMP1 SCG2 HTATSF1 DBH PCSK1N
intermediate	COL5A2 PTPRZ1 POSTN QKI PLP1 ZEB2 EDNRB ALDH1A1 COL1A1 MPZ	COL1A1 POSTN ZEB2 VCAN COL5A2 NEFL OLFML2A ALDH1A1 PTPRZ1 PLP1
precursors	S100B VIM METRN PLP1 SPARC S100A10 ANXA2 EDNRB MPZ IFITM3	PLP1 PTPRZ1 SPARC EDNRB ZEB2 VIM COL5A2 S100B MPZ QKI
sympathoblasts	PTMA STMN2 MAP1B PHOX2A GAL RTN1 GAP43 HAND2 HAND2-AS1 UCHL1	STMN2 RTN1 MAP1B GAL BASP1 GAP43 PHOX2A CD24 PTMA UCHL1

Genes of the model (97)

Data/data_train.h5ad, alphabetical order.

ACTG1	EIF4A2	NEFL	RPL41
ALDH1A1	FAU	NPY	RPS13
ANXA2	FOS	NRXN1	RPS2
APOE	GAL	OLFML2A	RPS29
ATP5F1E	GAP43	PCSK1N	RPS4X
B2M	GAPDH	PENK	RTN1
BASP1	H2AFZ	PHOX2A	S100A10
BEX1	HAND2	PLP1	S100A6
BTF3	HAND2-AS1	PNMT	S100B
C4orf48	HINT1	POSTN	SCG2
CARTPT	HMGB2	PPIA	SPARC
CD24	HMGN2	PRPH	STMN2
CHGA	HNRNPA1	PTMA	TH
CHGB	HTATSF1	PTMS	TIMP3
CNTN1	IFITM3	PTN	TMSB10
COL18A1	IGFBP2	PTPRZ1	TPM1
COL1A1	ITM2B	QKI	UBC
COL1A2	LGALS1	RAMP1	UCHL1
COL5A2	MAP1B	RGS4	VCAN
CRYAB	MDK	RGS5	VIM
DBH	MEG3	RPL10	YBX1
DLK1	MEG8	RPL12	ZEB2
EDNRB	METRN	RPL30	
EEF1A2	MPZ	RPL32	
EEF2	MT-CYB	RPL34	

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as $NB(k, c / s_i)$ when the depth factor s_i is applied.

Depth

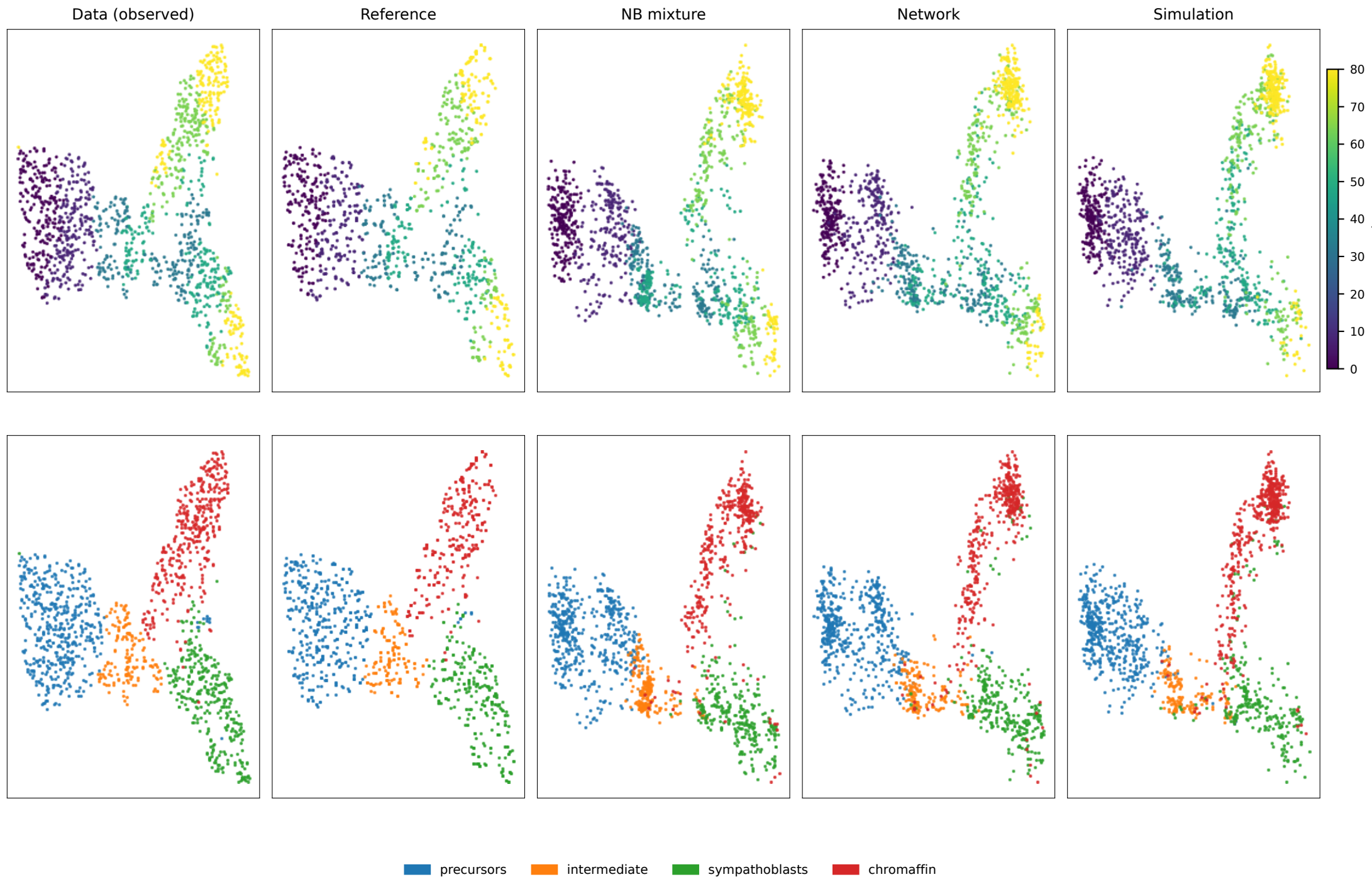
not estimated

Reason

97 genes < 10000

1. Generative model — trajectories and simulation

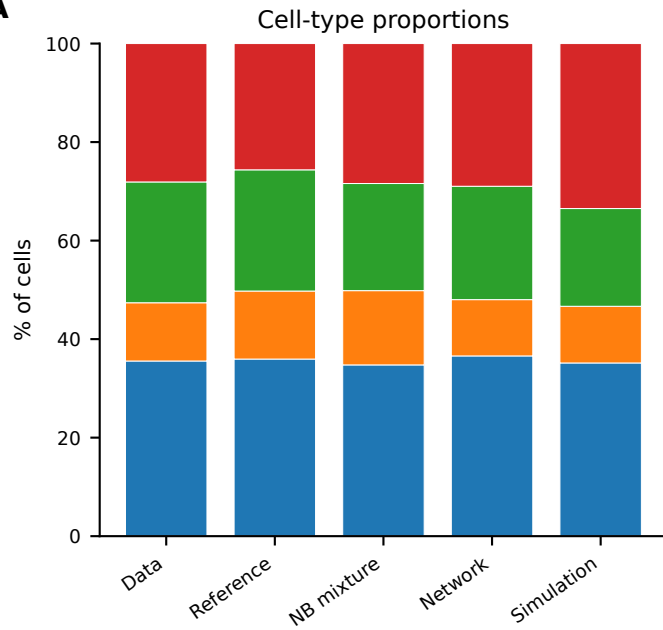
UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. without proliferation, as the simulation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



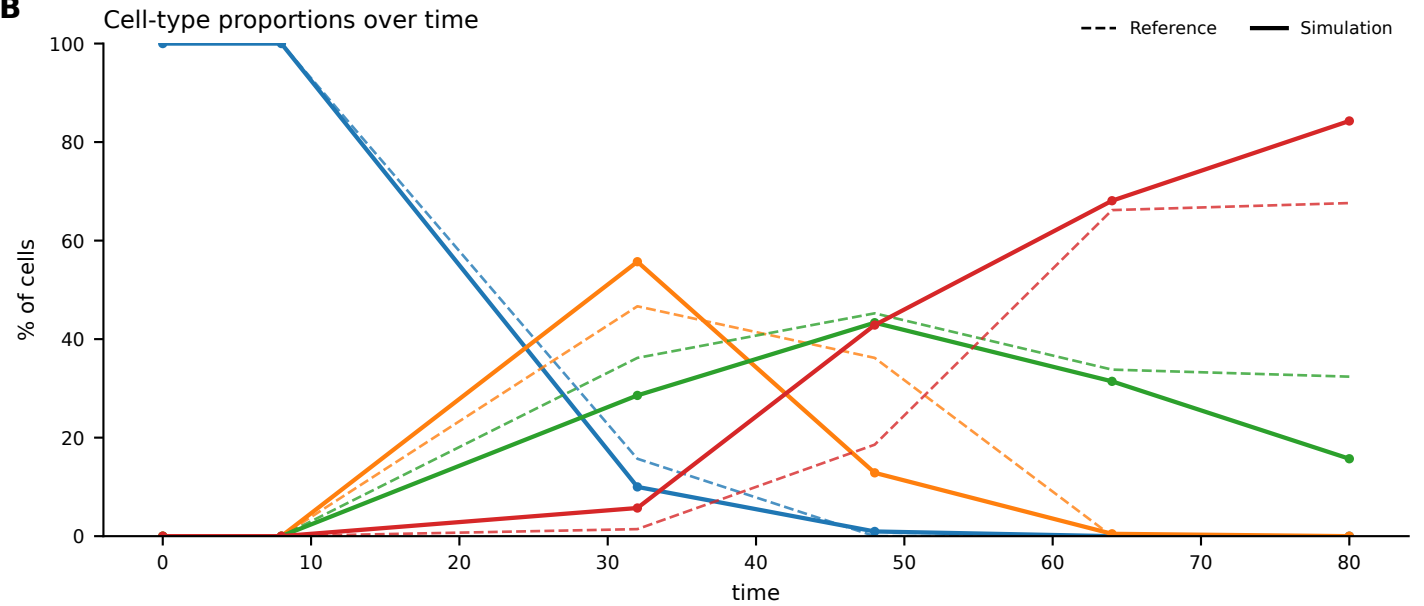
1. Generative model — quantitative checks

Reference = RNA of the trajectories equivalent to the simulation (without proliferation).

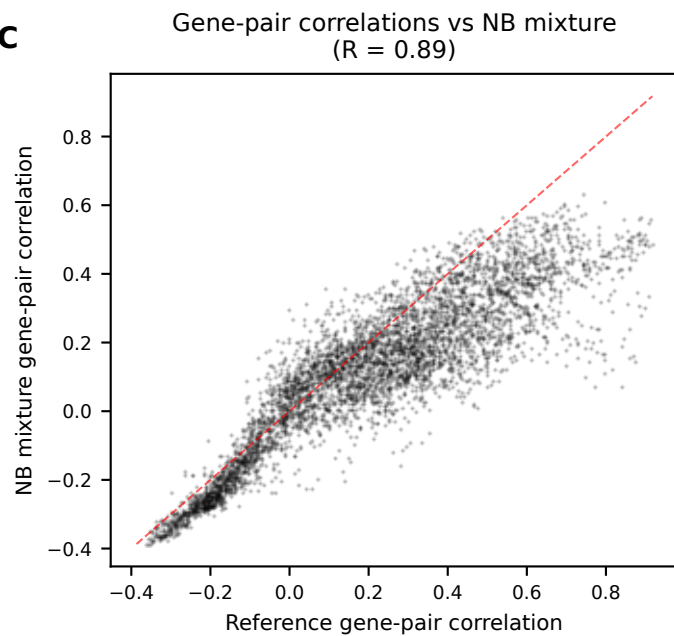
A



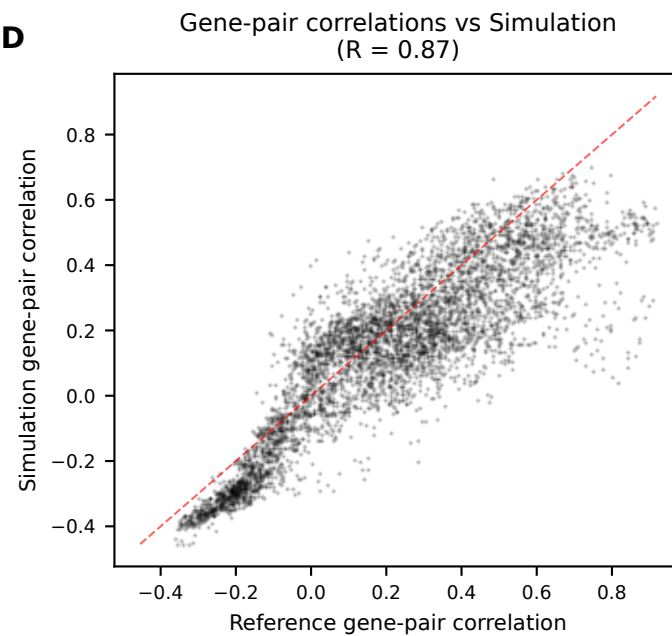
B



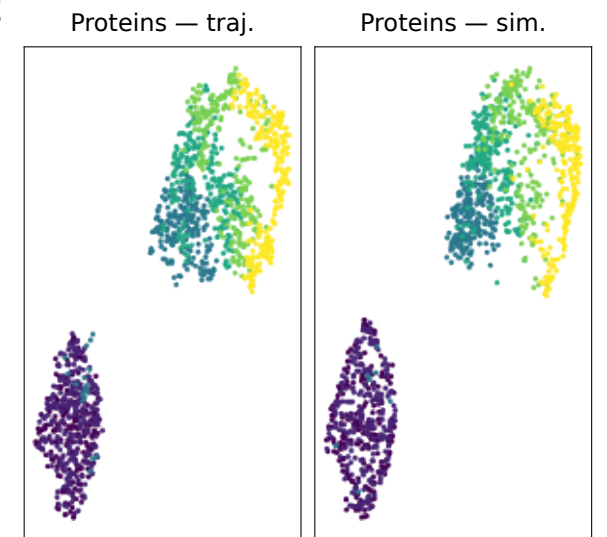
C



D



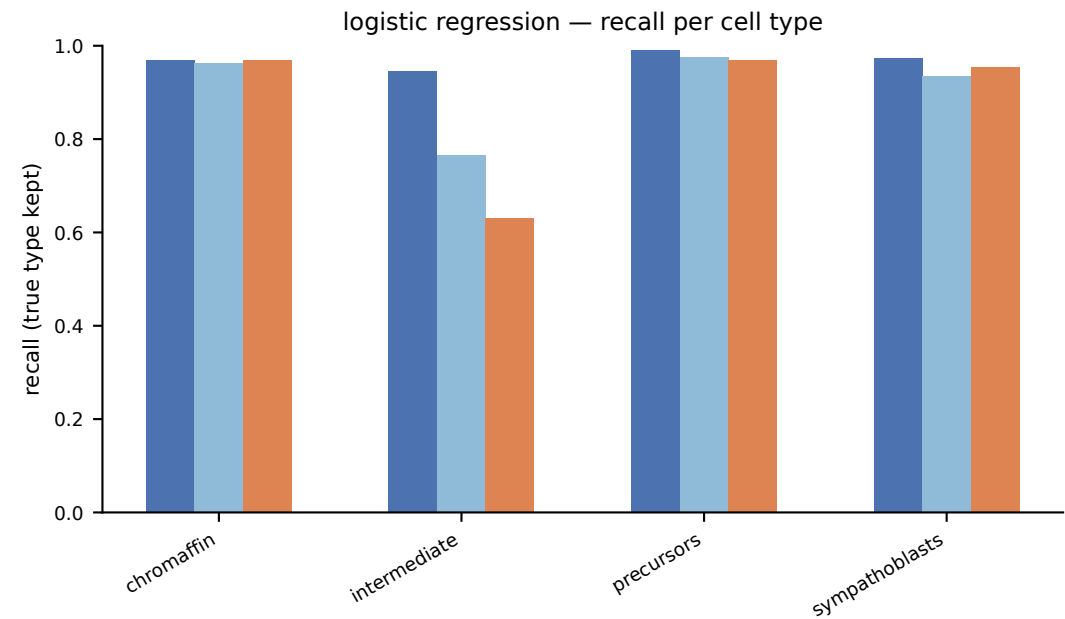
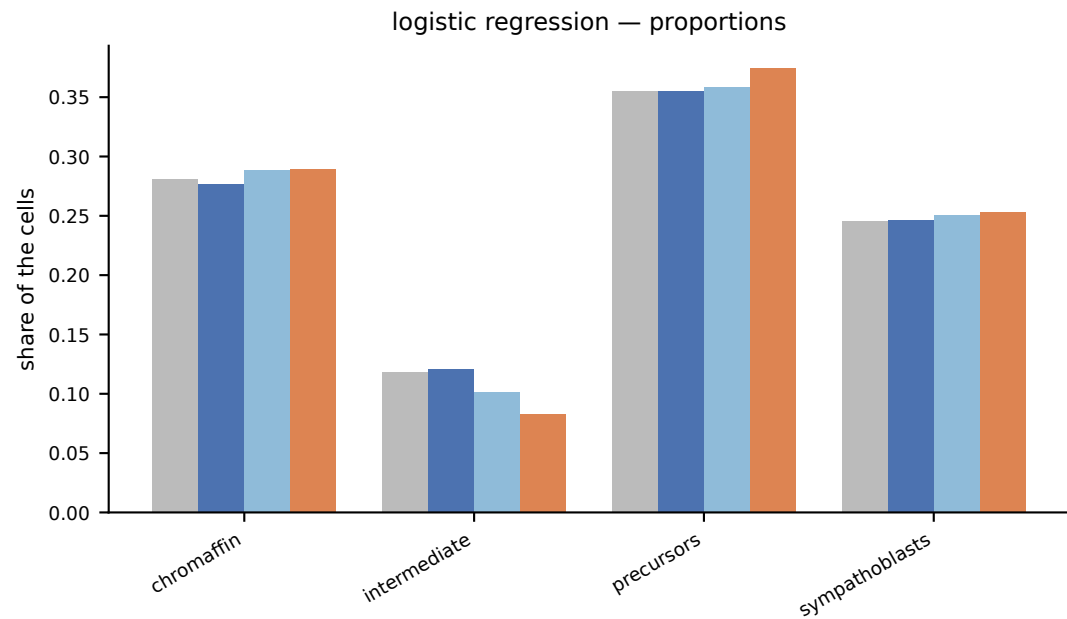
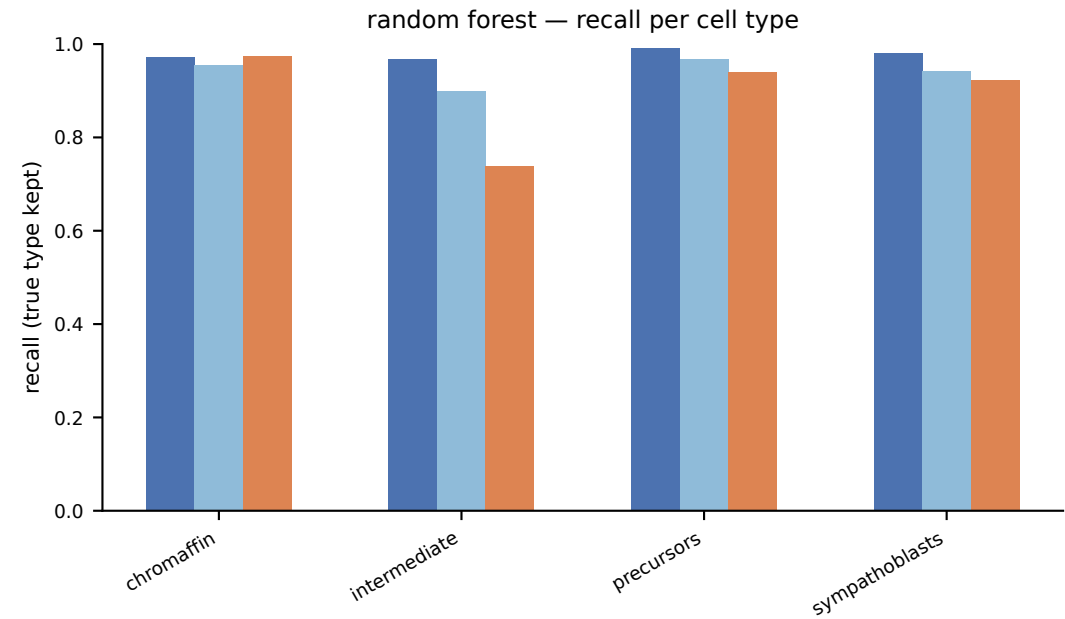
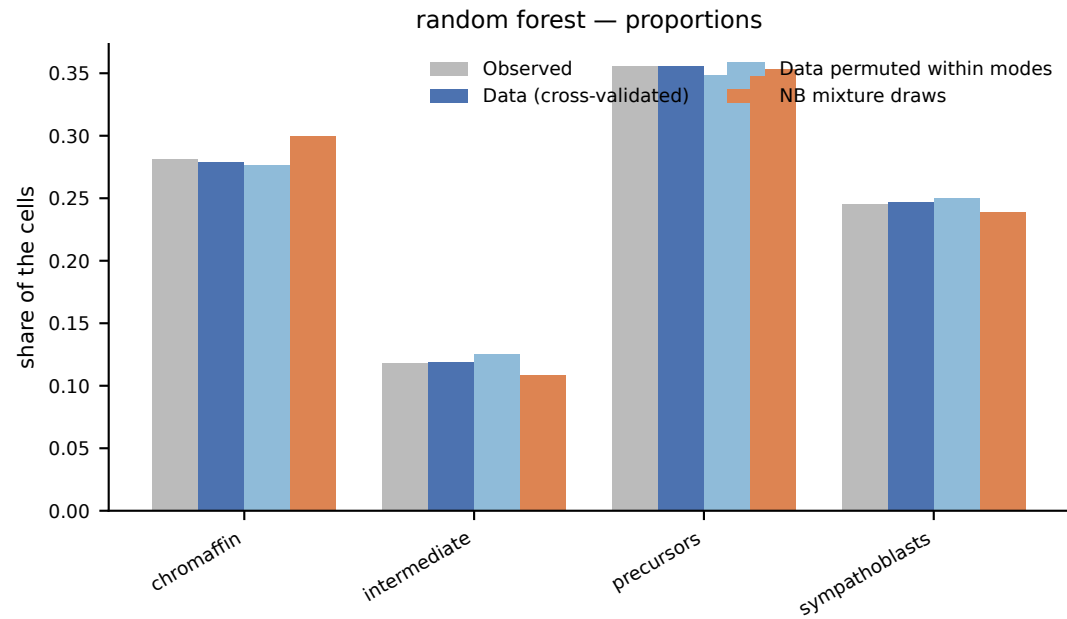
E



precursors intermediate sympathoblasts chromaffin

1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.

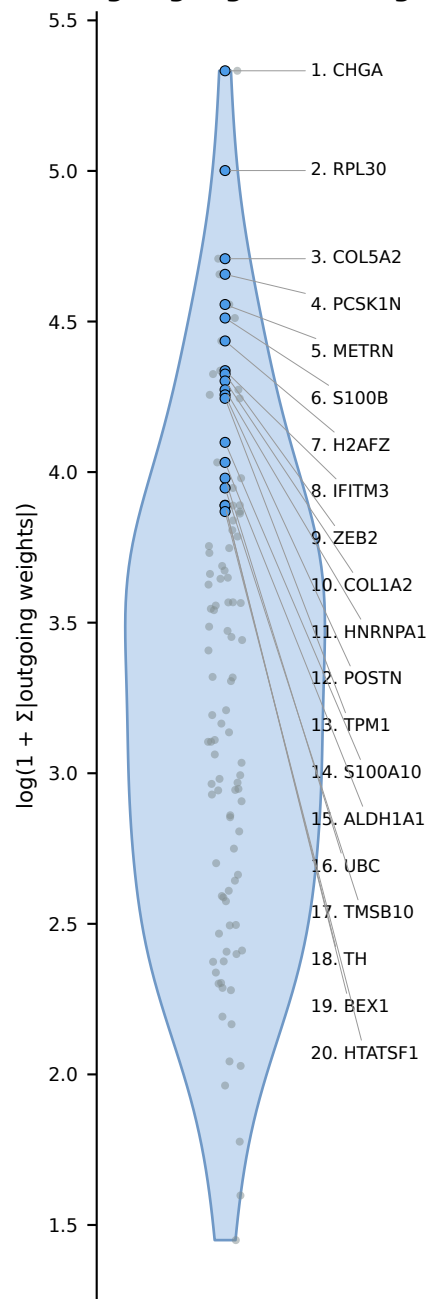


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus not shown (stimulus = 0.2 < 0.5).

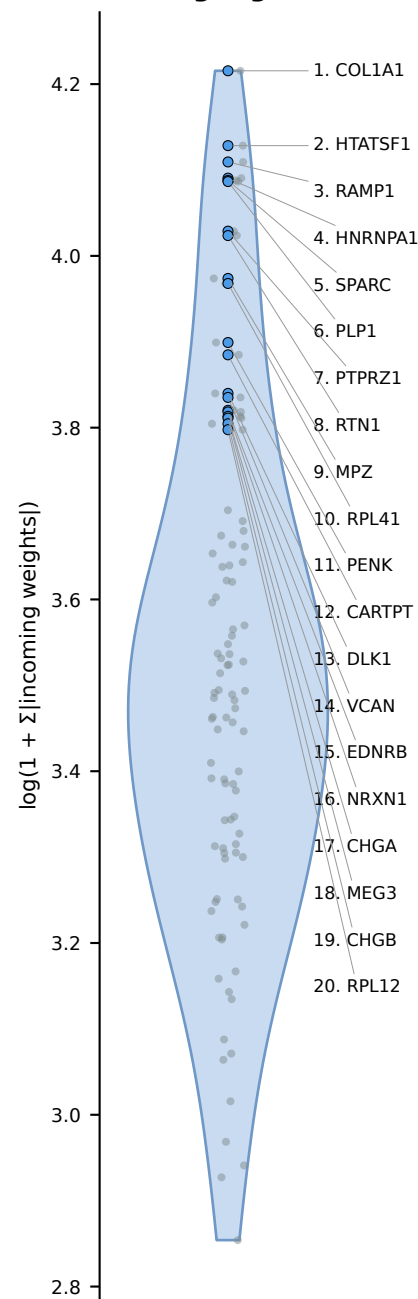
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C

Top 20 regulators

#	Regulator	$\Sigma w $	act.	inh.
1	CHGA	205.97	59	8
2	RPL30	147.60	70	18
3	COL5A2	109.89	1	52
4	PCSK1N	104.28	36	13
5	METRN	94.25	22	8
6	S100B	90.06	39	16
7	H2AFZ	83.39	54	20
8	IFITM3	75.47	20	16
9	ZEB2	74.60	3	42
10	COL1A2	72.89	20	26
11	HNRNPA1	70.77	25	34
12	POSTN	69.56	9	47
13	TPM1	68.74	43	45
14	S100A10	59.26	29	12
15	ALDH1A1	55.38	8	56
16	UBC	52.50	42	16
17	TMSB10	50.79	46	17
18	TH	47.91	28	23
19	BEX1	47.86	32	22
20	HTATSF1	46.89	24	6

D

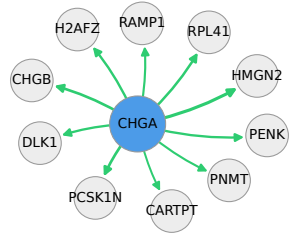
Top 20 regulated genes

#	Target	$\Sigma w $	act.	inh.
1	COL1A1	66.73	14	22
2	HTATSF1	61.08	13	29
3	RAMP1	59.91	27	35
4	HNRNPA1	58.77	28	26
5	SPARC	58.61	23	6
6	PLP1	58.53	19	8
7	PTPRZ1	55.19	17	33
8	RTN1	54.91	15	33
9	MPZ	52.18	19	27
10	RPL41	51.87	25	28
11	PENK	48.36	23	43
12	CARTPT	47.66	19	37
13	DLK1	45.52	11	36
14	VCAN	45.30	43	29
15	EDNRB	44.61	21	20
16	NRXN1	44.53	43	25
17	CHGA	44.30	16	37
18	MEG3	44.20	9	36
19	CHGB	43.91	19	31
20	RPL12	43.60	26	13

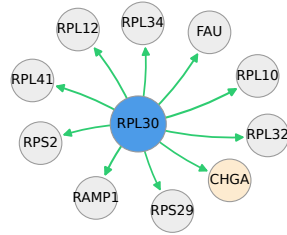
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest ($|w|$) targets of the regulator AND the 10 strongest regulators of the target; edge width $\propto |weight|$ (global scale).
 Perturbed genes (KO/OV) are highlighted in beige.

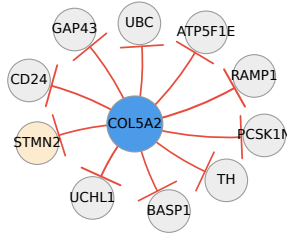
#1 CHGA



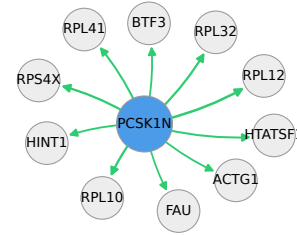
#2 RPL30



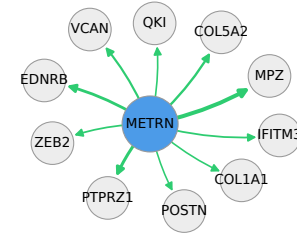
#3 COL5A2



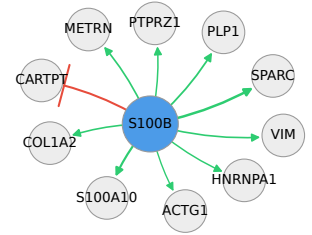
#4 PCSK1N



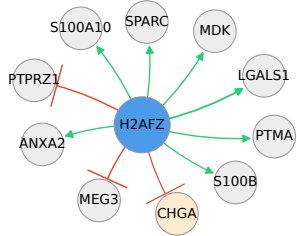
#5 METRN



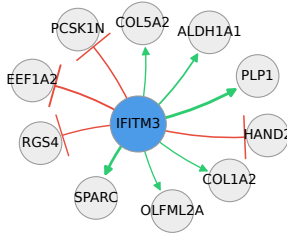
#6 S100B



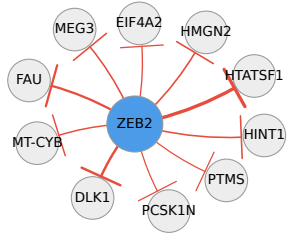
#7 H2AFZ



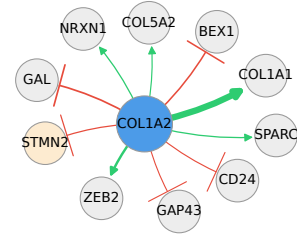
#8 IFITM3



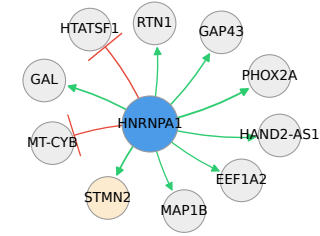
#9 ZEB2



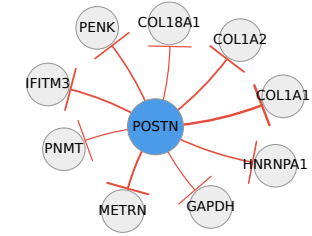
#10 COL1A2



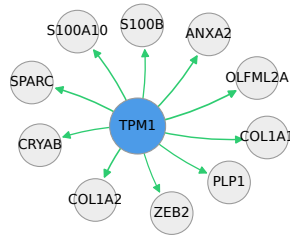
#11 HNRNPA1



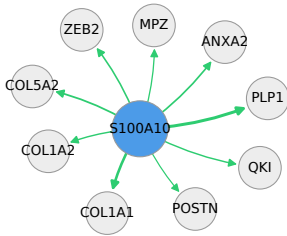
#12 POSTN



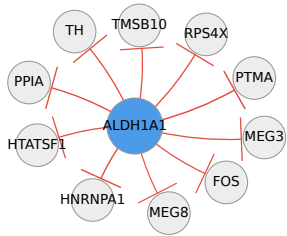
#13 TPM1



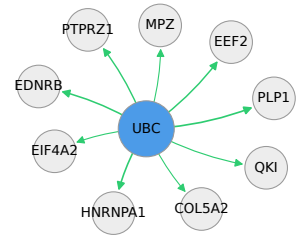
#14 S100A10



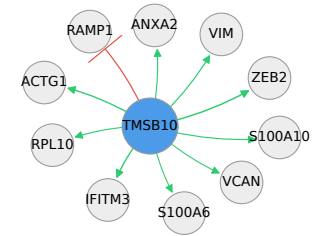
#15 ALDH1A1



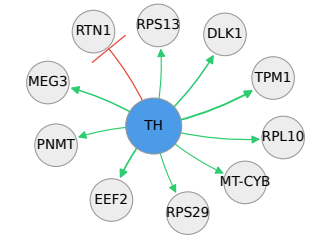
#16 UBC



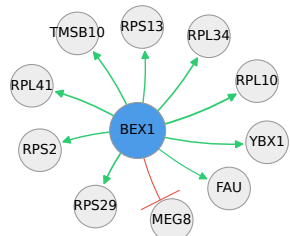
#17 TMSB10



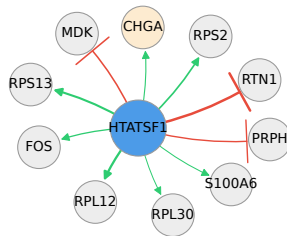
#18 TH



#19 BEX1



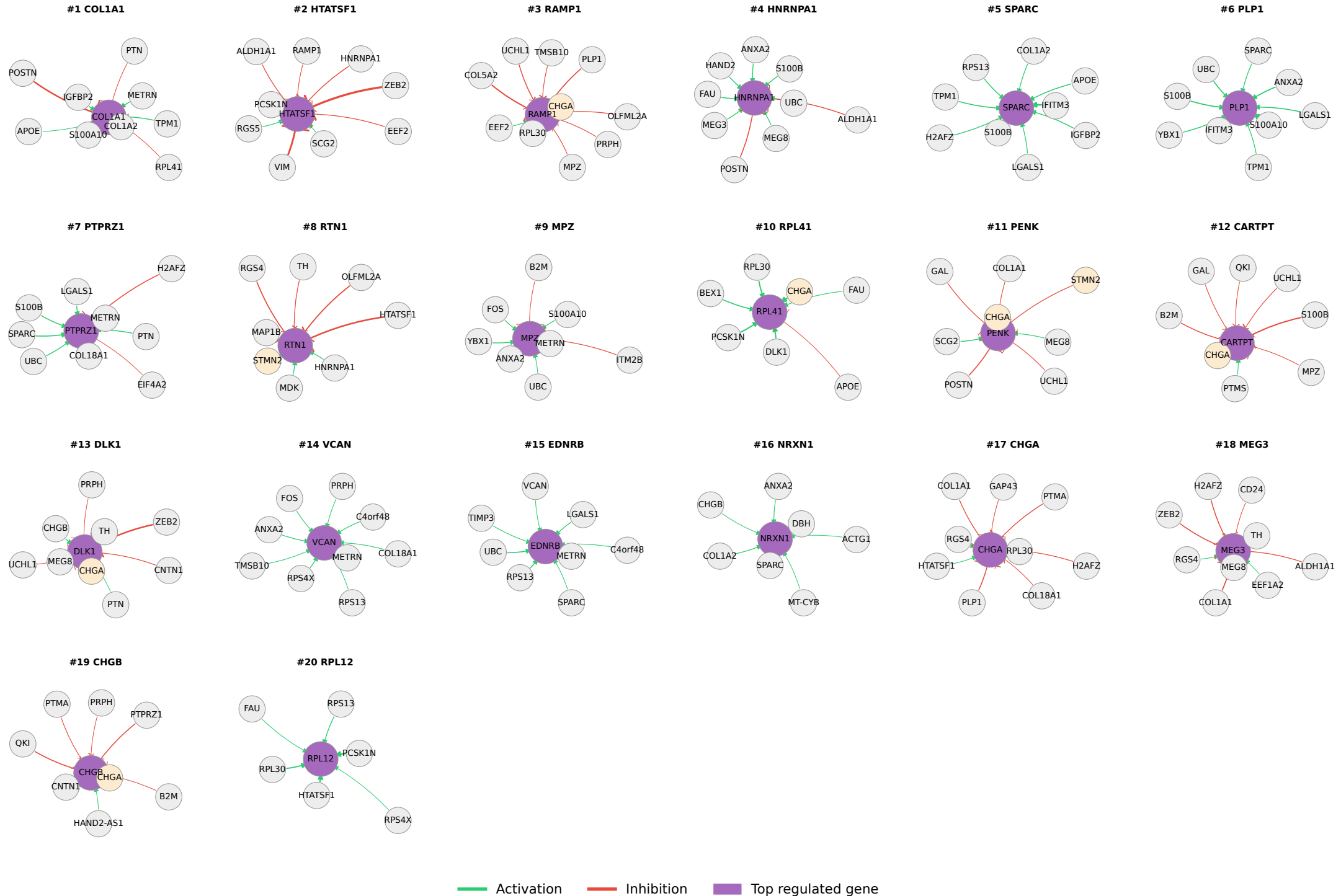
#20 HTATSF1



— Activation — Inhibition — Top regulator

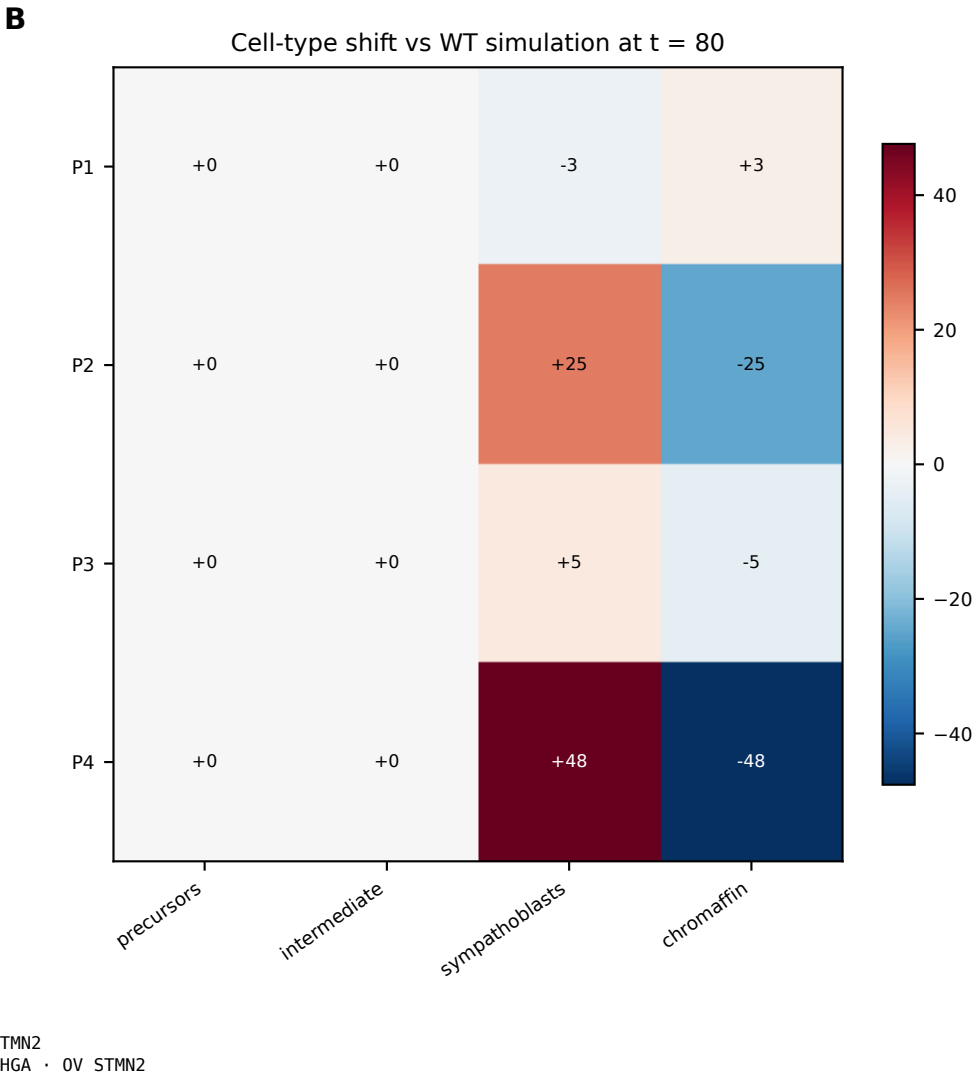
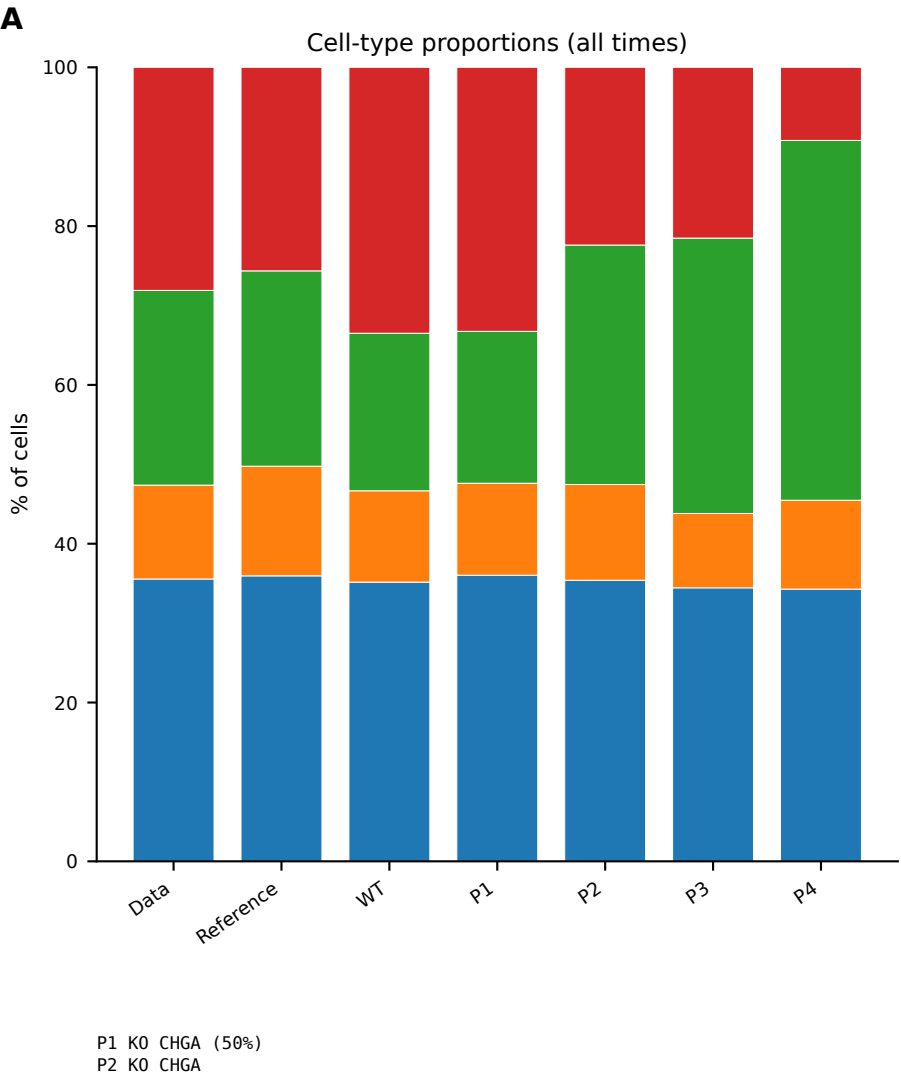
2. Gene regulatory network — top regulated genes and their main regulators

Edges shown only if among the 10 strongest ($|w|$) targets of the regulator AND the 10 strongest regulators of the target; edge width $\propto |weight|$ (global scale).
 Perturbed genes (KO/OV) are highlighted in beige.



3. In-silico perturbations — overview

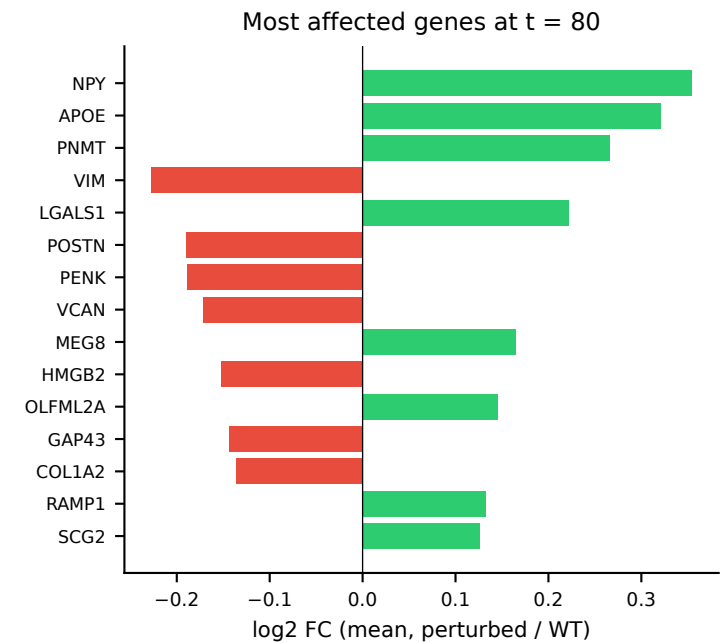
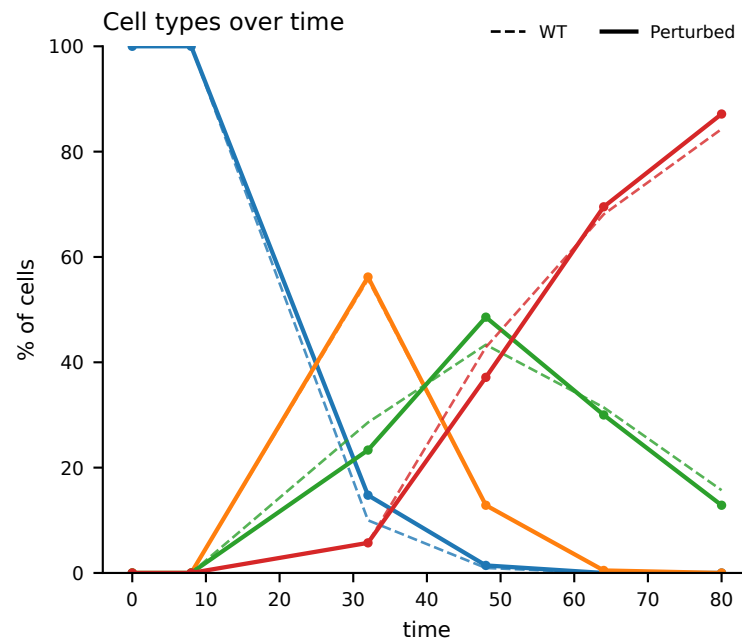
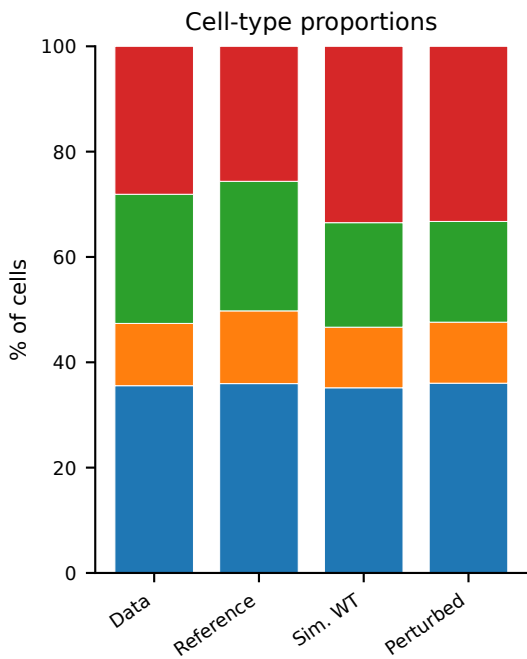
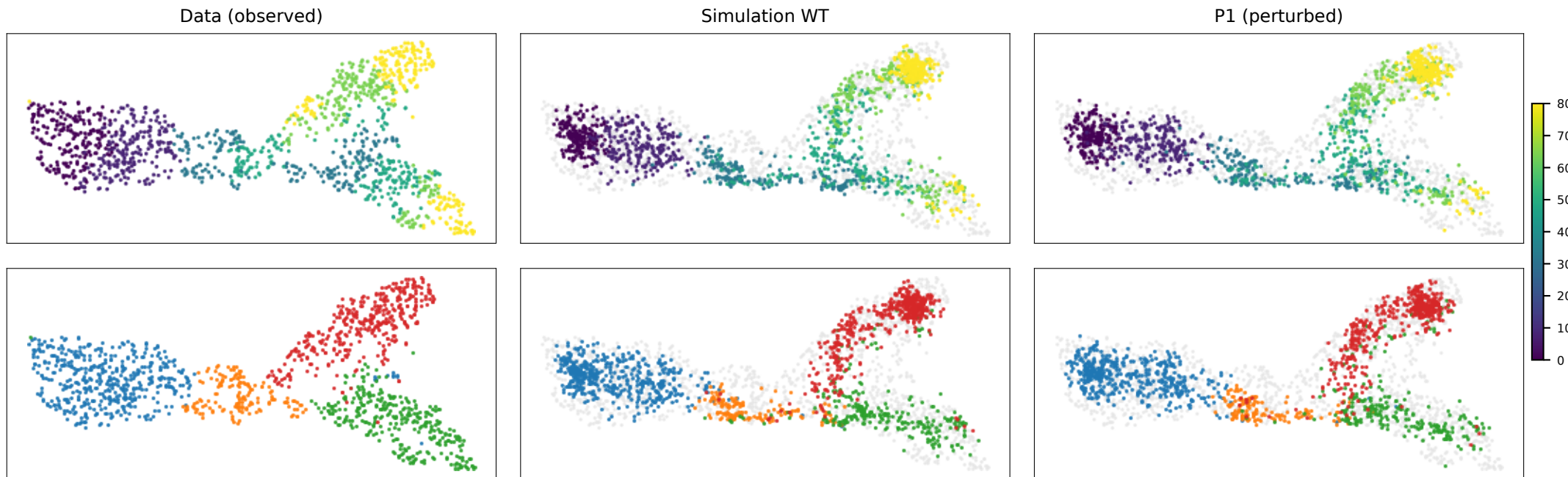
Cell types predicted by per-sample classifiers (logistic) trained on the observed data.



precursors intermediate sympathoblasts chromaffin

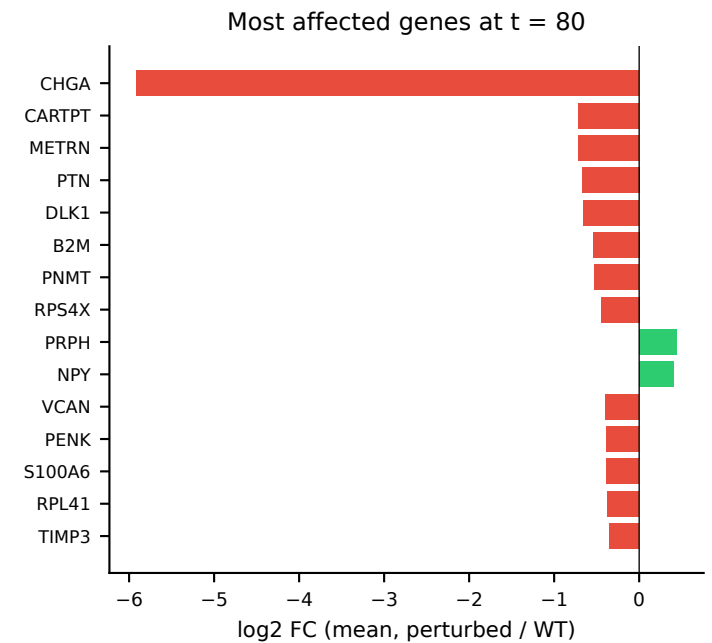
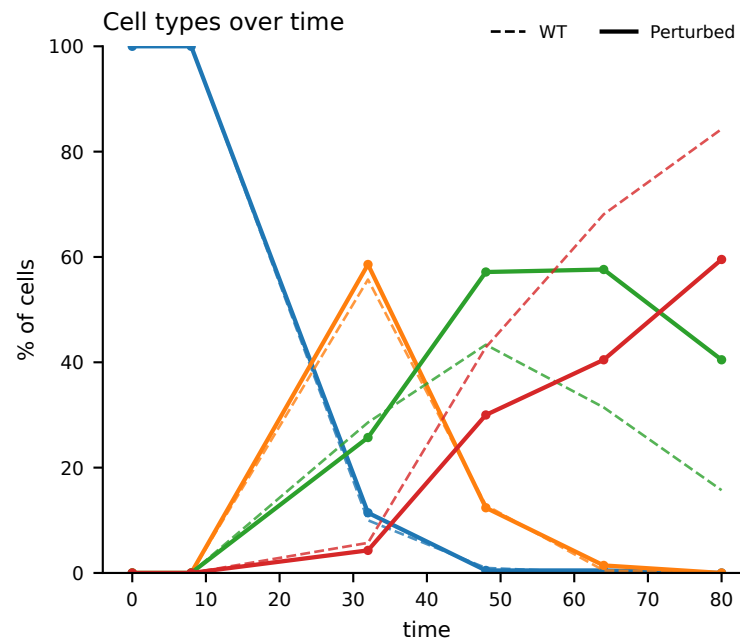
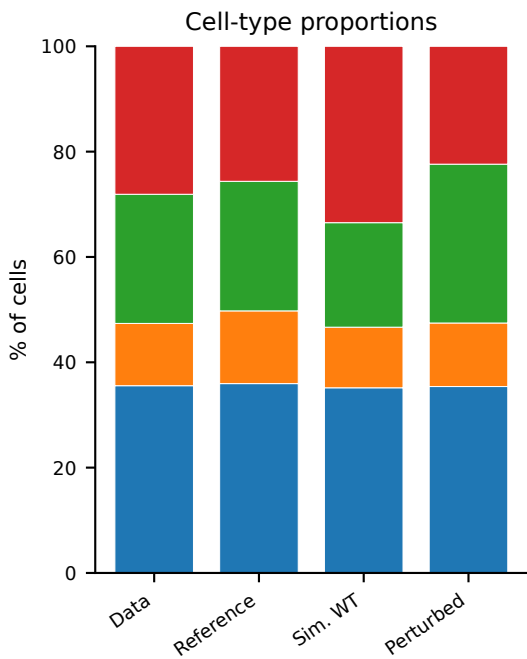
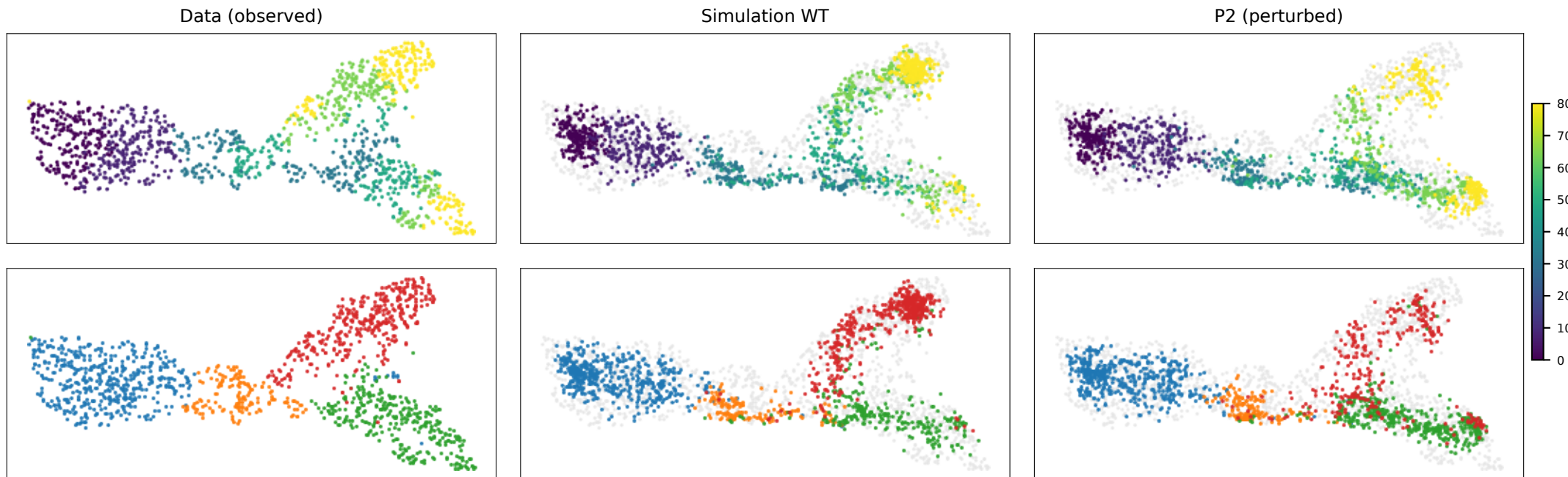
3. Perturbation P1 — KO CHGA (50%)

KO_CHGApct50_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



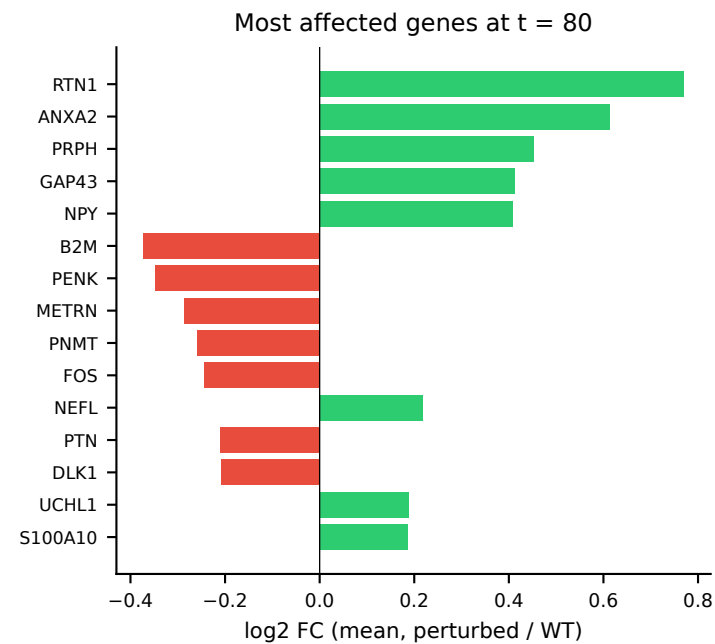
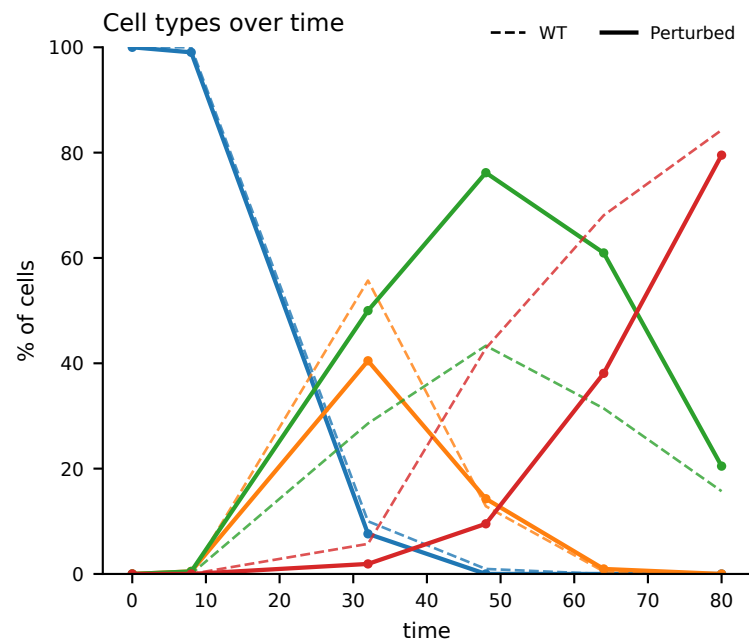
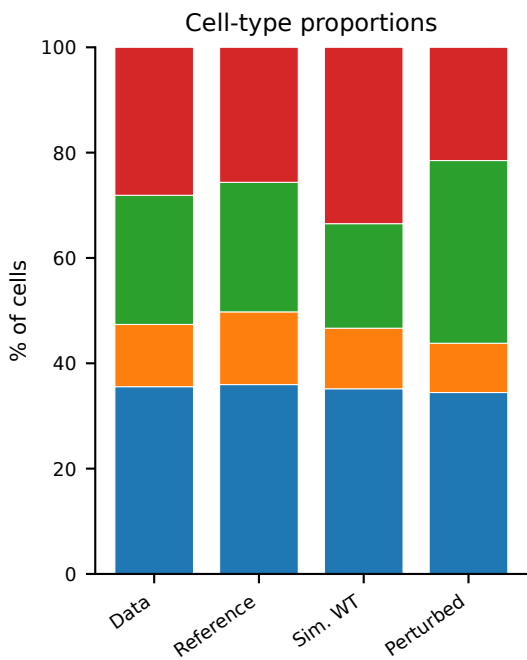
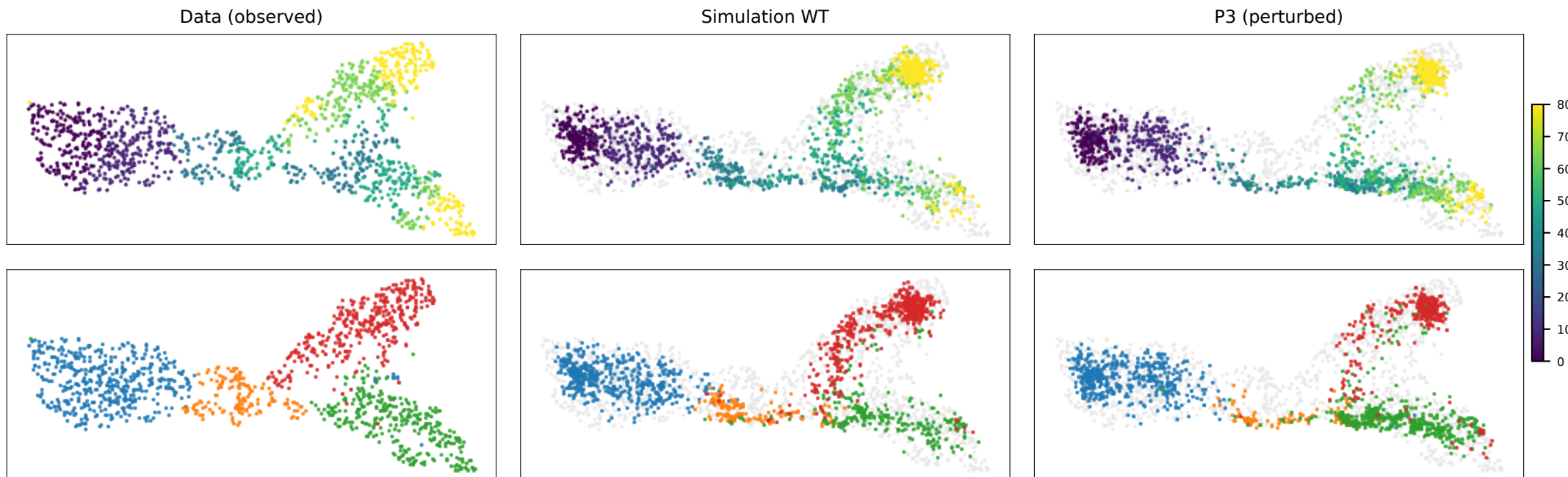
3. Perturbation P2 — KO CHGA

KO_CHGA_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



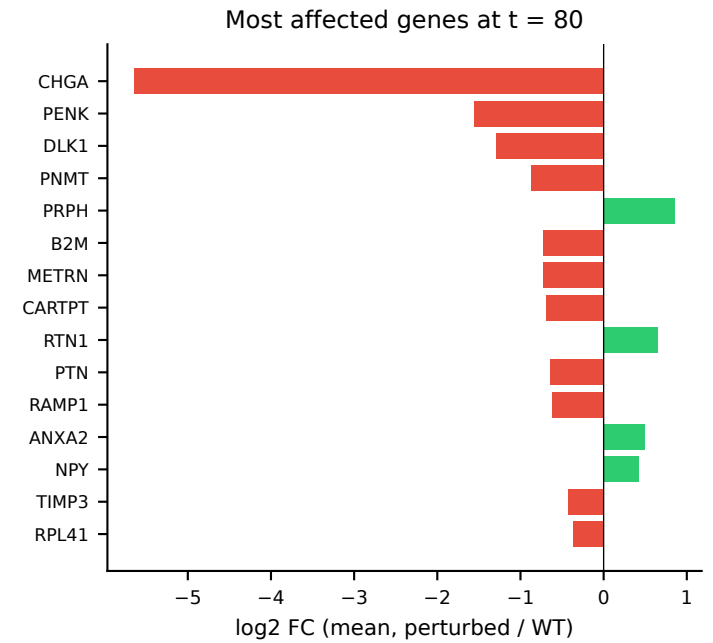
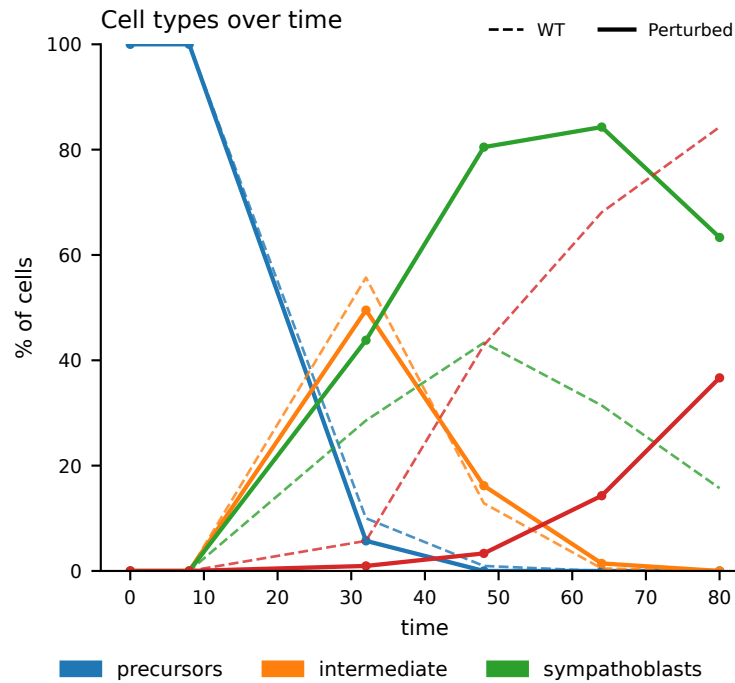
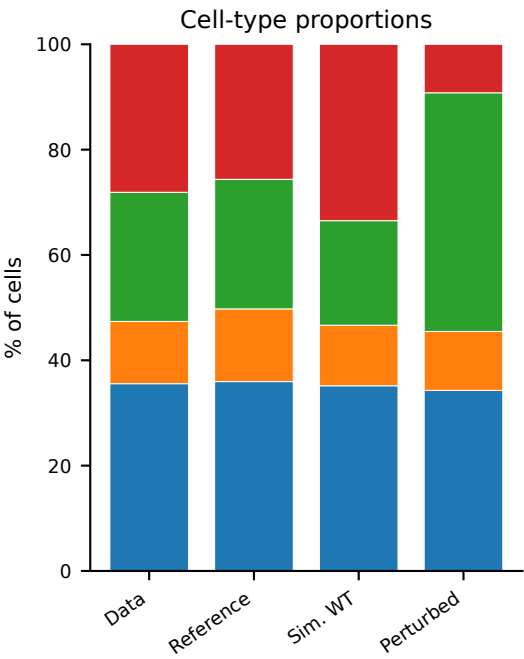
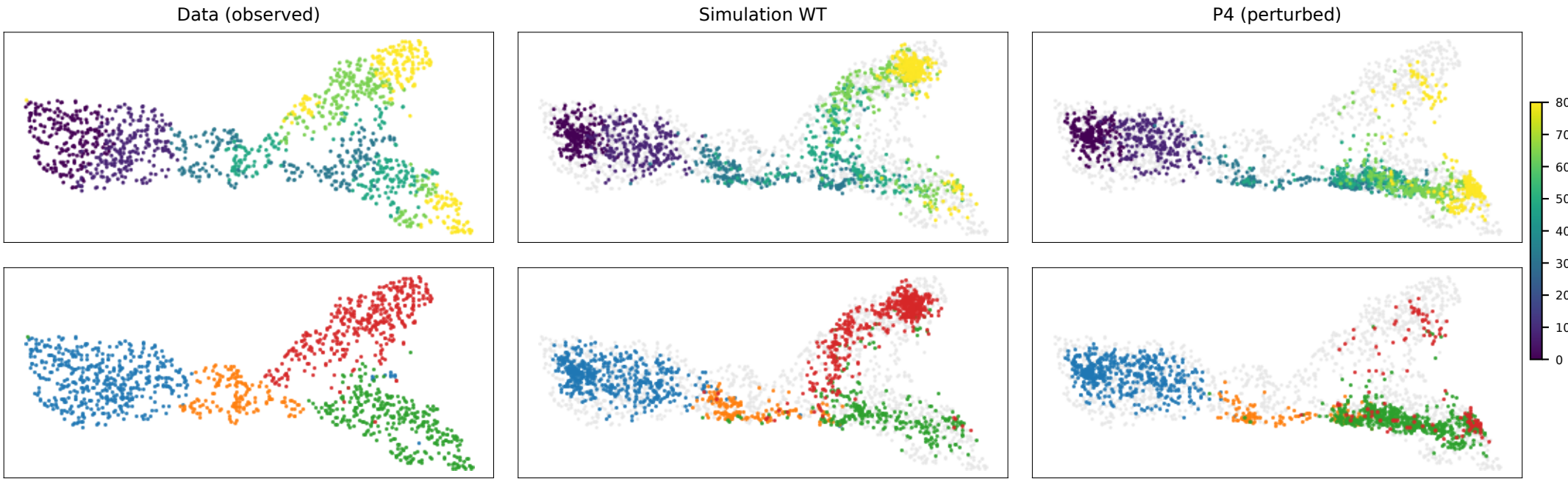
3. Perturbation P3 — OV STMN2

KO_none_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



3. Perturbation P4 — KO CHGA · OV STMN2

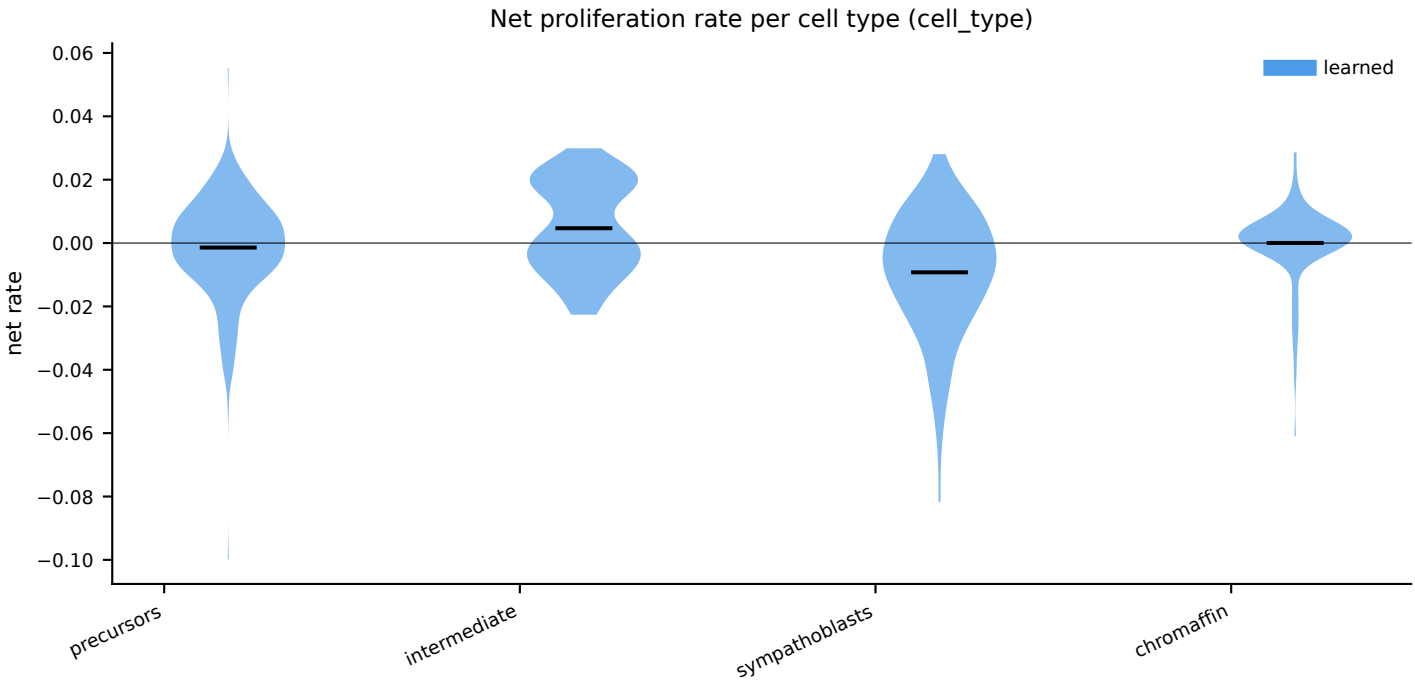
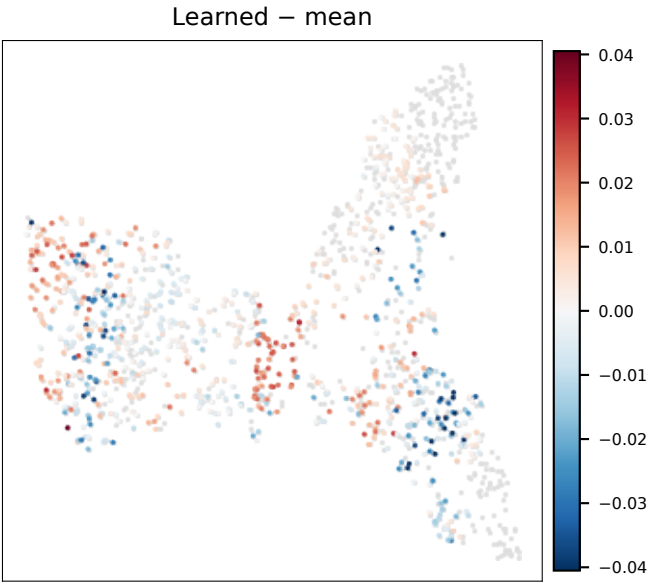
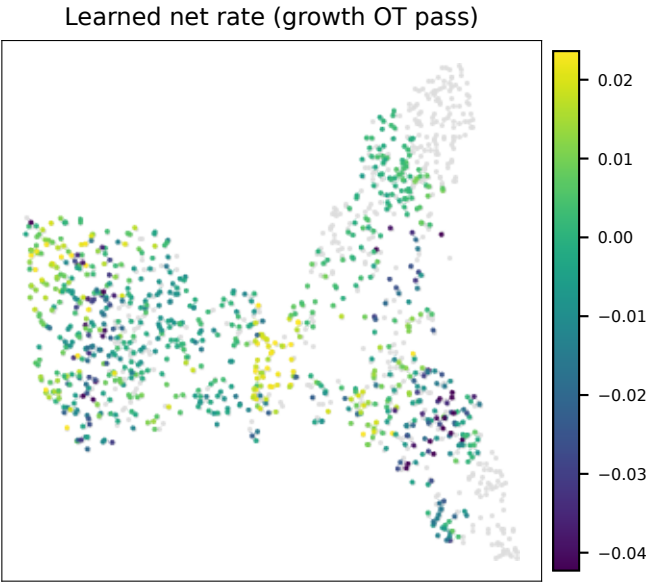
KO_CHGA_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



4. Proliferation — prior vs learned net rate (birth – death)

Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

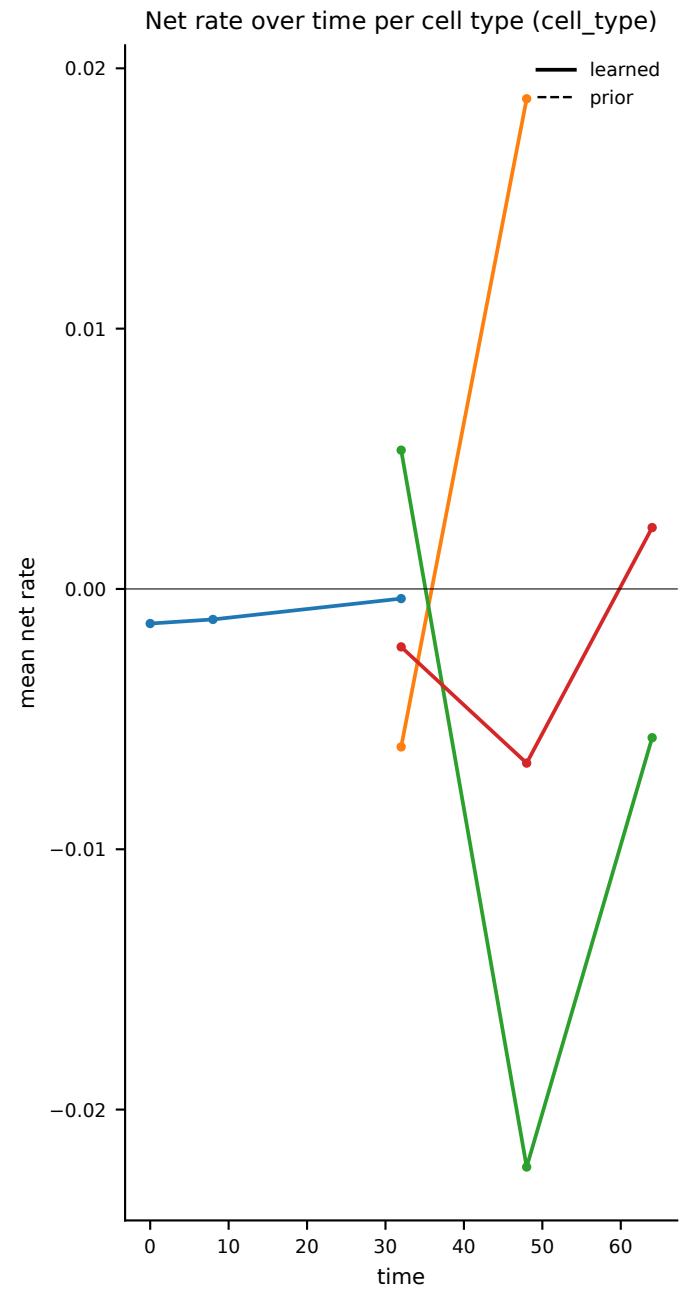
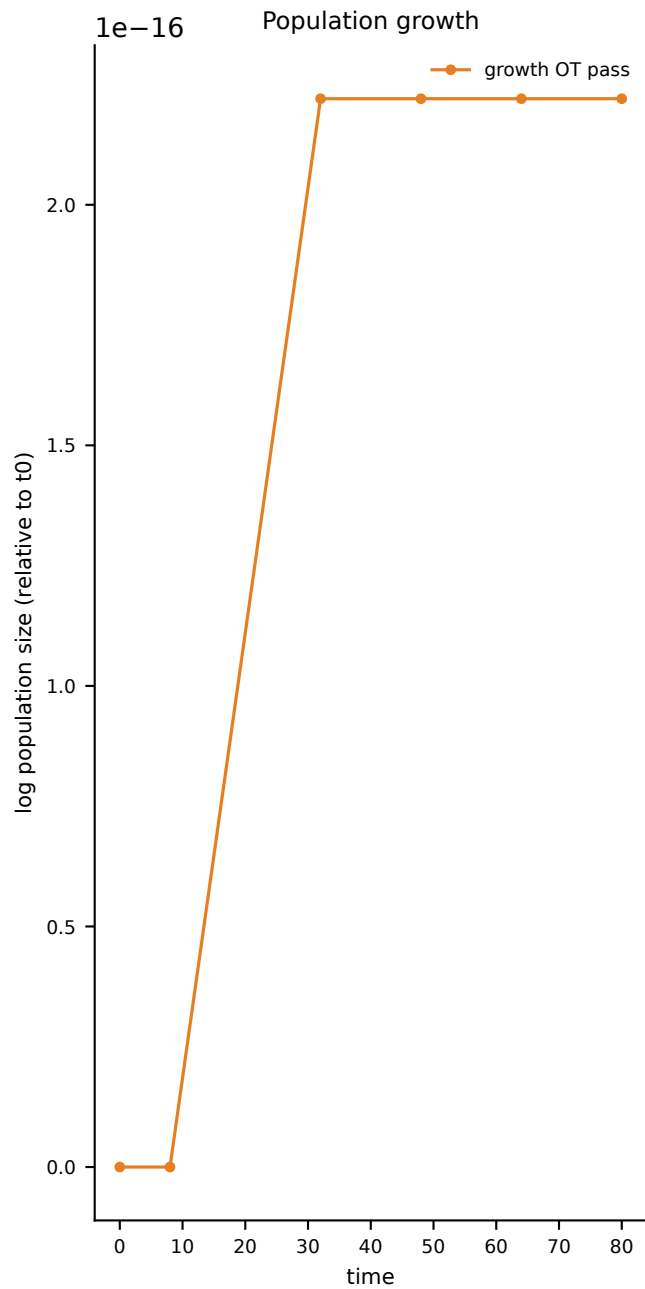


Mean net rate per cell type (cell_type)

Cell type	Anchor	Prior	Learned
precursors	—	—	-0.00142
intermediate	—	—	0.00468
sympathoblasts	—	—	-0.00924
chromaffin	—	—	3.25e-05

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; unless the sheet population_sizes is filled, its absolute level follows the prior.

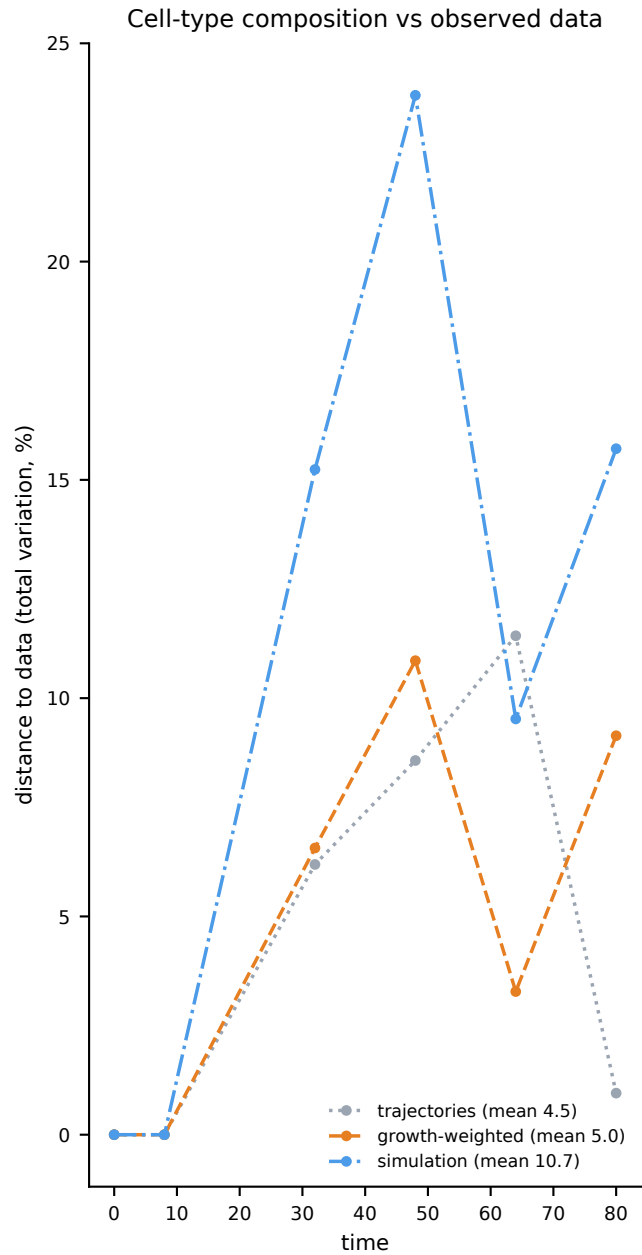
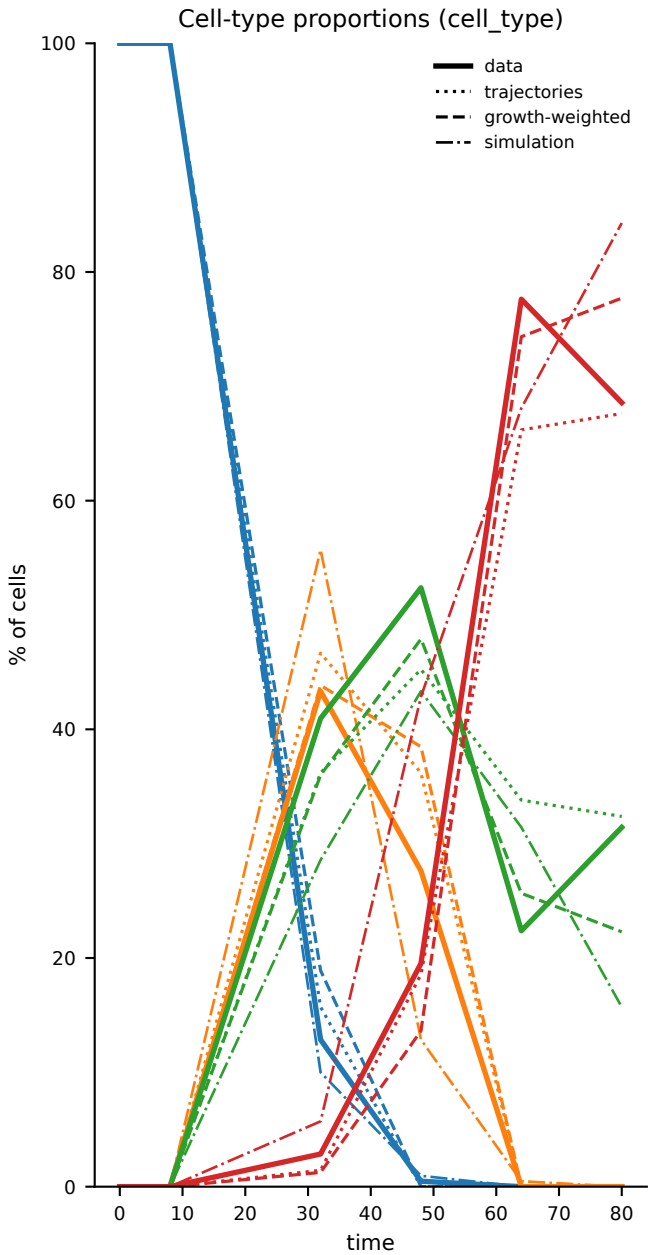


No proliferation MLP
(simulate_with_proliferation = True,
then infer_network_simul)

precursors intermediate sympathoblasts chromaffin

4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion $\exp(JR_{\text{opt}})$ should bring them back to the observed data.



No MLP fit diagnostics
(rerun infer_network_simul
with proliferation)

precursors intermediate sympathoblasts chromaffin

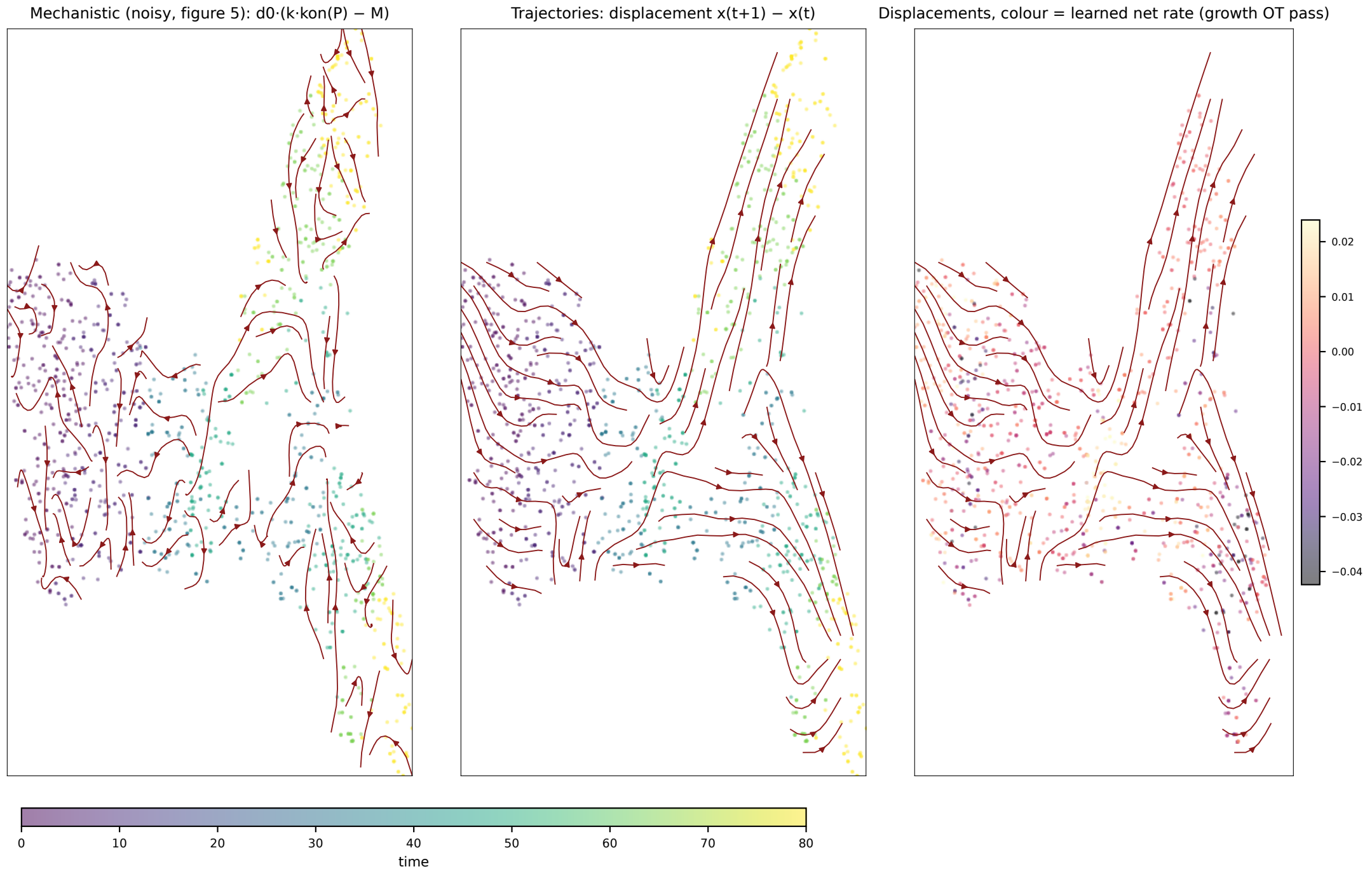
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

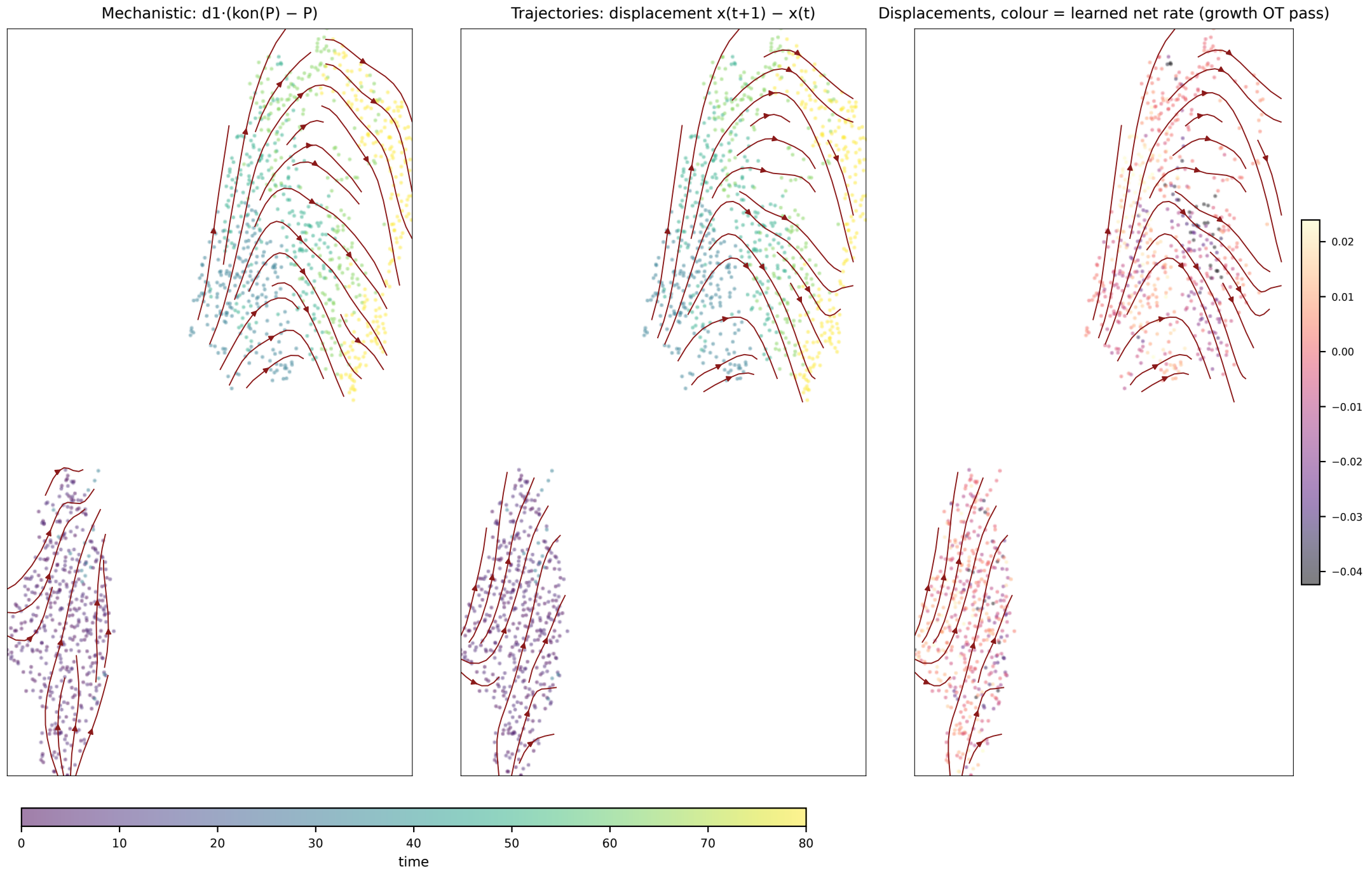
5. Learned dynamics — mRNA velocity and displacement fields

All 1260 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.03.



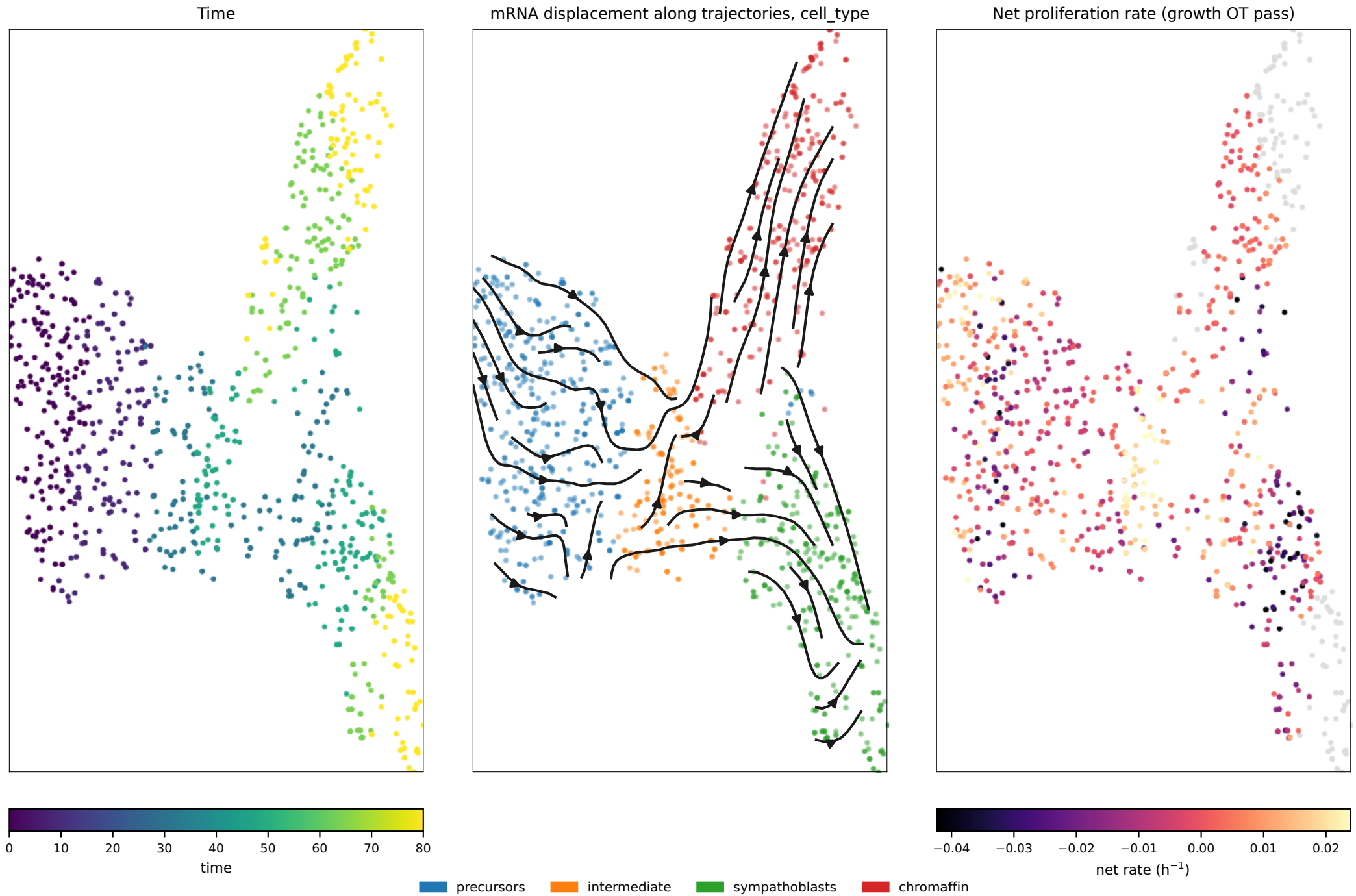
5. Learned dynamics — Proteins velocity and displacement fields

All 1260 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.53.



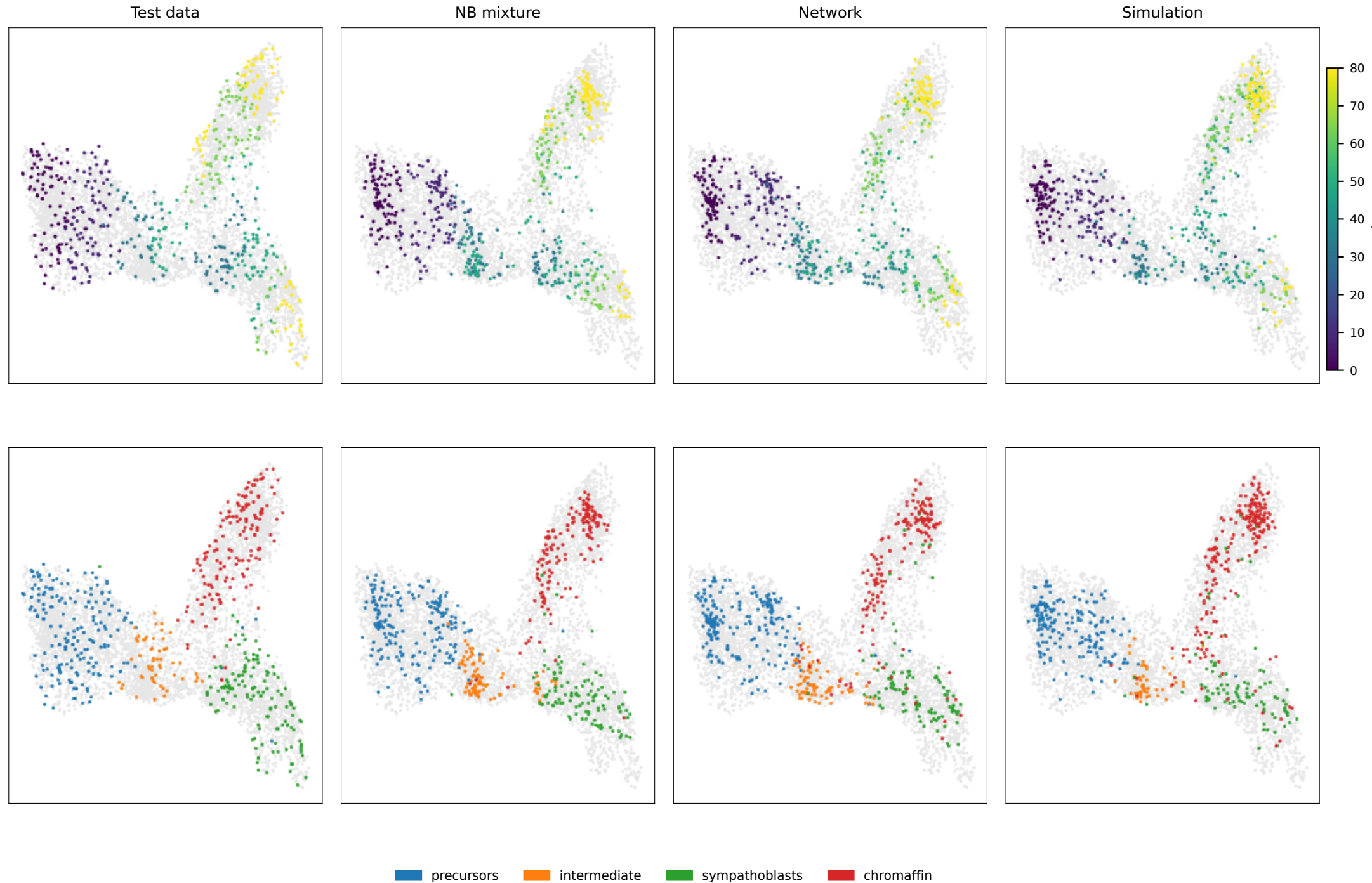
5. Learned dynamics — summary

All 1260 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories $x(t+1) - x(t)$; background of the field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

